

Large language Models-empowered automatic knowledge graph development based on multi-modal data for building health resilience

Tianlong Shan^{a,b}, Fan Zhang^{a,*}, Albert P.C. Chan^a, Shiyao Zhu^c, Kaijian Li^b

^a Department of Building and Real Estate, The Hong Kong Polytechnic University, Hong Kong SAR, China

^b School of Management Science and Real Estate, Chongqing University, Chongqing 400045, China

^c Department of Wood Science, The University of British Columbia, Vancouver, Canada V6T 1Z4

ARTICLE INFO

Keywords:

Building health resilience
Knowledge graph
Large language models
Multi-modal data
Rainstorm

ABSTRACT

Improving the health resilience of building (BHR) helps keep stable health status of both the building and its occupants under disasters. As BHR is an emerging concept, there is no structured knowledge graph to understand the whole process of BHR under disasters. Therefore, this study aims to build a structured BHR knowledge graph based on multi-modal data, providing sufficient structured knowledge for BHR enhancement. An automated knowledge graph construction approach is proposed to empower the ontology design and triple extraction by large language models (LLMs), and validation processes based on In-context Learning (ICL) prompts. A case study is conducted to construct the knowledge graph of BHR under rainstorms in Hong Kong. The performance of the proposed LLMs-empowered knowledge extraction is also validated based on natural language processing metrics and LLMs-based Evaluation (LLMs-Eval). BHR knowledge graph indicates the potential relations between disasters, factors, response actions, and the health status of the building and occupants, and provides insight to guide the BHR enhancement. The superiority of the proposed LLMs-empowered automated knowledge graph construction approach is proven, implying LLMs have great potential in knowledge graph construction, not only for BHR but also for other concepts that require structured knowledge for further explorations and analyses.

1. Introduction

The health of occupants is usually tied to the health status of the buildings they inhabit or work in. The presence of numerous disasters, such as rainstorms, typhoons, earthquakes, and epidemics, has numerous detrimental effects on the health of the occupants, subsequently impacting their welfare [1]. Being static containers of human activities, buildings and their health status are receiving increasing attention [2]. Buildings with better health resilience indicate that buildings have a strong capability to resist disasters' impact on their health status and recover to normal health status rapidly, reducing the impacts on occupants as much as possible. "Health resilience of building" (BHR) is proposed to describe this capability [3].

In practice, the response process of BHR is complex and it encompasses various knowledge across multiple aspects. For instance, the occupants' behaviors, the building's attributes, the internal and external environment of the building, the actions taken in response, and other dimensions can be influencing factors of BHR [4,5]. This complex and

diversified knowledge implies that proposing BHR management and enhancement strategies is challenging. Thus, there is an urgent need for a method to systematically organize and integrate the structured knowledge related to BHR, which can provide insight into different knowledge dimensions and establish effective control over BHR [6].

Since BHR is a novel concept without sufficient structured data and relevant literature [5], the analysis of BHR response process and ontological concepts is lacking, posing a significant obstacle to the construction of BHR knowledge graphs. Additionally, most of existing knowledge graph research used structured textual data to construct knowledge, overlooking multi-modal information beyond the text data, such as images and videos. Multi-modal data can provide an insightful knowledge base from visual and textual levels, which can play a crucial role in the structuring process of knowledge graphs. In terms of knowledge graph construction methods, current techniques often require complex manual annotation, and even some deep learning algorithms need pre-annotation. This can result in the knowledge graph being heavily influenced by subjective factors and complex manual

* Corresponding author.

E-mail addresses: stlong@stu.cqu.edu.cn (T. Shan), fan-2.zhang@polyu.edu.hk (F. Zhang), albert.chan@polyu.edu.hk (A.P.C. Chan), syzhu@mail.ubc.ca (S. Zhu), likaijian@cqu.edu.cn (K. Li).

<https://doi.org/10.1016/j.aei.2025.103655>

Received 31 December 2024; Received in revised form 5 July 2025; Accepted 10 July 2025

Available online 16 July 2025

1474-0346/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

annotation work [7]. Methods that use pre-trained models for named entity recognition and relationship identification without manual annotation also have limitations, as these methods are only applicable to specific concepts such as addresses, names, and dates.

Large Language Models (LLMs), with their robust natural language processing and data analysis capabilities, have been applied by numerous scholars to various urban issues [8–11]. However, LLMs remain unstudied in areas related to urban resilience, especially in the management and enhancement strategies of building health resilience. This lack of research highlights a significant gap in existing knowledge, considering the crucial role of building health resilience in ensuring urban sustainability.

To address these limitations, this study uses LLMs to empower the BHR knowledge graph construction process based on multi-modal data. Multi-modal data, including text, images, and videos, contains rich information that can address the data issues for the BHR knowledge graph. Ontology is foundational in constructing a knowledge graph [12]. This study first designs an ontology for BHR adopting a combined approach of top-down and bottom-up methods. Second, since LLMs have exceptional natural language processing (NLP) capabilities and can provide profound insights into the dynamic context of language [13], ChatGPT-4o (one of the LLMs) is applied to propose a novel named entities and relations recognition method for triples extraction. Feedback checking mechanism (FCM) and in-context learning (ICL) prompts are integrated into the triple extraction process to ensure accuracy. The BHR under rainstorms in Hong Kong is chosen as a case study to construct knowledge graph based on multi-modal data from media, to further provide decision support for BHR enhancement strategies under rainstorms.

Three main contributions have been made in this study. 1) This study proposes an LLMs-empowered automated method for constructing and validating BHR knowledge graphs. This method includes ICL-based prompt engineering and the FCM to ensure the accuracy of the knowledge graph construction, and it holds significant value for other research that needs to construct knowledge graph automatically. 2) This study innovatively incorporates multi-modal data into the knowledge graph construction process and utilizes LLMs' capabilities for processing. The multi-modal data will have more extensive applications in constructing knowledge graphs with the continuous upgrade of generative AI products. 3) This study constructs a knowledge graph for BHR under rainstorms in Hong Kong, which can facilitate understanding and exploration of the response process of BHR under rainstorms, and guide to enhance BHR levels in Hong Kong.

2. Literature review

This section primarily forms a comprehensive overview by reviewing historical scholars' perspectives on concepts related to BHR, methods for ontology construction, and methods for constructing knowledge graphs.

2.1. Building health resilience

Resilience is a key concept in the pursuit of sustainability, and it is also an essential topic of research in sustainable development [14]. Resilience was initially used to describe the stability of ecosystems, which refers to the ability of ecosystems to absorb disturbances and changes, while still maintaining the relationships between populations or state variables within the ecosystem [15]. Urban resilience refers to the ability of the city as a system to resist internal chaos or external disturbances like earthquakes, floods, climate change, etc. [16]. A city is a large system composed of many smaller subsystems, such as buildings, infrastructure, and social environment, and includes organizations made up of people, such as governments, communities, etc. The assessment system of urban resilience varies with the different disturbances faced by the city, and the evaluation index system needs to establish a comprehensive and scientific evaluation system based on the characteristics and consequences of the interference [17–19].

Overall, the scope of urban resilience is quite broad, making it challenging to propose targeted and effective strategies for enhancing resilience. As a result, some scholars have begun to further study the subsystems of urban resilience, such as transportation system resilience [20], community resilience [21], and building resilience [22]. Building resilience is an emerging topic because buildings are the most critical components of a city, and the building subsystem also includes both the physical structures and the resident elements. Existing research on resilience at the building level primarily includes seismic resilience, thermal resilience, and fire resilience [23–25]. However, there is a notable lack of studies on resilience from the “health” perspective. The health status of buildings needs more attention because buildings themselves have characteristics like structural damage, repair, and reconstruction, and the occupants have psychological and physiological health features. Additionally, people spend approximately 90 % of their time inside buildings [26], which is why BHR is gradually receiving more attention from scholars [3,27]. The BHR studied in this article is essentially a microcosm of urban resilience, with the resilience subject narrowed down from the city to a single building, and the measurement of resilience status is focused on the health status. To measure BHR, Zhang proposed measuring the condition of building health to represent BHR through the typical function loss fluctuation curves of building groups and validated it with actual cases from Hong Kong public housing [3]. The influencing factors of BHR are also studied by combining Generative AI's predictive model and topic word model to extract topic words from unstructured text data [5].

The concept of BHR is essentially an intersection of urban resilience, health resilience, and building health. Urban resilience addresses a broader range of dimensions, encompassing various aspects such as urban economy, society, environment, and architecture. Research on urban resilience plays a significant role in urban disaster governance. In contrast, “Building Health Resilience”, which places greater emphasis on the health status of buildings and their occupants, is also decisive in enhancing the “Health status” of buildings. There are plenty of existing studies related to urban resilience, but the limited research on BHR lacks a thorough exploration and clear understanding of the concept's intrinsic meaning and the interconnections. Therefore, there is a need for research aimed at establishing structured knowledge to advance a comprehensive understanding and management of BHR. This entails creating a detailed framework or model that delineates the components of BHR, identifies key indicators, and illustrates the dynamics between various elements within the response process.

2.2. Resilience ontology design

The concept of ontology generally refers to a formal and standardized expression of a shared conceptual model [28]. There are numerous studies on ontology construction, with the primary methods based on a top-down approach [29] of field abstraction and conceptual division, and a bottom-up [30] construction method that maps structured data resources to knowledge graphs based on rules.

Indeed, some studies have employed ontology construction to explore universally prevalent themes and unified concepts in resilience. In the field of supply chains, due to the uncertainties and various types of failures in the manufacturing supply process, supply chain resilience ontology is constructed using web language rules and semantic methods to form a knowledge base for supply chain networks, which can be used to resist the impact of these uncertainties [31] and also promote automating supply chain decision-making. In the field of food systems, resilience ontology was constructed to explore the relationships and mutual mappings between various concepts and classifications in the food system [32].

Current scholars designing resilience ontology all mentioned that resilience is a rich concept, and it varies across different fields subjects. This is rooted in the cognitive impact of different disciplines. Therefore, in the process of constructing the ontology of BHR, it is necessary to start

from the connotation of the concept of building health, combine the corpus, semantics, and other aspects in the field to absorb insights and incorporate them into the BHR construction process in this study.

2.3. Knowledge graph construction

Knowledge graphs are an emerging technology based on the visualization of network relationships of thematic knowledge, with wide applications in semantic parsing, question-answer system development and more [33]. The modeling and topological structure of knowledge graphs can clearly demonstrate the relationships between various entities, playing a significant role in guiding practice. The construction process of a knowledge graph generally includes the following aspects: ontology construction, named entity recognition (NER), entity relationship identification, knowledge fusion, and knowledge storage [12]. Knowledge fusion integrates entities and relationships that express the same concept differently based on NER, reducing the complexity of the knowledge graph. Knowledge storage involves storing the constructed knowledge graph in the form of a database.

The methods for extracting entities and relations include methods such as automatic recognition based on manual annotation, pre-trained models, and semi-automatic recognition [34]. Early manual annotation was based on rules or knowledge methods, such as BIO (B-Begin, I-Inner, O-Others) and BMES (B-Begin, M-Middle, E-End, S-Single) or BIOSE, etc. The advantage of this method is the presence of a specific field's knowledge dictionary, which results in high-precision entity extraction [35]. However, the downside is the general lack of such dictionaries in new fields, coupled with the high cost of manual annotation. Supervised learning approaches have been proposed to address the issues of manual annotation. Common machine learning models include SVM (Support Vector Machine), HMM (Hidden Markov Model), and CRF (Conditional Random Fields). Bi-LSTM (Bidirectional Long Short-Term Memory) networks are capable of learning bidirectional long-term dependencies in time-series data while retaining past and future information at any given point [36]. Supplementing the Bi-LSTM with a CRF layer improves the accuracy of label prediction and decoding. Furthermore, to obtain more precise text-embedded vectors, the BERT (Bidirectional Encoder Representations from Transformers) model is also used in the entity recognition process, making the BERT-BiLSTM-CRF model a mainstream model currently. The superiority of the BERT-BiLSTM-CRF model over previous entity recognition models has been proven in research on knowledge graphs concerning electronic interference in railway accidents in China [37]. Table 1 summarizes various knowledge graphs and their corresponding entity-relationship identification methods, and this table is essential for understanding the landscape of knowledge graph construction.

The disadvantages of manual and semi-automatic recognition lie in the need for a large amount of complex manual labor for extensive entity annotation. The scope of automatic recognition based on pre-trained models is limited. For instance, some pre-trained models offer automated extraction of concrete entities such as people, time, location, and events [46].

Therefore, there is currently a lack of a method that can automatically recognize entity relationships for knowledge graphs in new concepts or fields. The necessity for such a framework is driven by the rapid expansion of information across various fields, necessitating tools that can quickly adapt and extract relevant knowledge without extensive manual intervention [47]. An ideal solution would not only reduce the time and resources required for building field-specific knowledge graphs but also enhance the accuracy and comprehensiveness of the extracted knowledge.

2.4. The application of large language models

LLMs have become the predominant technology in generative artificial intelligence, exhibiting robust capabilities in generation, transfer,

Table 1

Review of knowledge graph construction methods and data sources.

Author	Topic	Annotation	Method of NER	Data source
[38]	Ship Collision Accident Knowledge Graph	BIO	TextRank	Reports
[39]	Construction Site Hazards Knowledge Graph	Labels	Computer Vision	Images
[40]	Railway Operational Accidents Knowledge Graph	Labels	Manual Labeling	Reports
[41]	Electric Power Equipment Faults Knowledge Graph	BIO	BERT-BiLSTM-CRF	Letter of Credits
[42]	Academic Knowledge Graph	BIO	BiLSTM-CRF	Blogs
[43]	Tourism Knowledge Graph	Text Classification	BERT-BiLSTM-CRF	Encyclopedia data
[44]	Accident Knowledge Graph	BMES	BERT-BiLSTM-CRF	Reports
[45]	Urban Rainstorm Disaster Event Knowledge Graph	BMES	BERT-BiLSTM-Attention-CRF	Reports

and user interaction [48,49]. These models are artificial intelligence systems built upon the Transformer deep learning framework, incorporating core technologies such as self-attention mechanisms, multimodal fusion, fine-tuning instruction, retrieval-augmented generation (RAG), and prompt engineering [50]. These technologies empower LLMs with generalization abilities, contextual learning, and emergent properties.

The advancement of LLM technology has facilitated the development of various products, which can generally be categorized into open-source and closed-source models. Closed-source models often function as core technologies for companies, contributing to a wide array of large model products in the market. Notable models include OpenAI's ChatGPT series, Google's Gemini 1.5 Pro, and Anthropic's Claude 3 series. Among these, the earliest and most recognized model is ChatGPT, based on the generative pre-trained transformer (GPT). This series has evolved from GPT-2 (1.5 billion parameters) in 2019, to ChatGPT-3.5 (175 billion parameters) in November 2022, and GPT-4 (300 billion parameters) in March 2023. ChatGPT-4o serves as the multimodal version of ChatGPT-4, capable of processing multimodal tasks, including text, images, and audio [51].

Due to their powerful natural language processing, learning, and interactive capabilities, LLM have been applied in research related to resilience. For example, Li employed prompt engineering with LLMs to assist in quantifying supply chain risks, thereby deriving measures to enhance supply chain resilience and inform enterprise strategies and decision-making [52]. In another study on the resilience of the global automotive supply chain network, Li utilized a method combining LLMs with a multi-layer perceptron (MLP) to quantify risk points within the network and forecast the resilience of non-direct suppliers [53]. This research demonstrated that models integrated with LLMs significantly outperformed other machine learning models. In the field of cybersecurity, LLMs have also been combined with traditional machine learning and deep learning models to quantitatively assess performance indicators in cybersecurity systems, which in turn has helped to strengthen

cybersecurity policies for small and medium-sized enterprises [54]. In addition, LLMs have also been widely applied at the urban level to address various city-related issues. For example, Feng et al. proposed the CityGPT framework to integrate diverse urban knowledge for solving urban spatial tasks [9]. Sun et al. combined GIS data, deep learning techniques, and GPT to analyze street-view images and architectural features [10]. GPT has also been used to develop the ontology for building performance simulation, enhancing prevention efficiency and value by integrating simulation engines and advanced data, thus promoting the development of building performance simulation systems [11]. Choi et al.'s research verified that ChatGPT can conduct data analysis, energy prediction, building feature engineering and optimization, as well as case analysis in Urban Building Energy Modeling (UBEM) tasks, offering more valuable insights to urban planners and policymakers [8].

Although LLMs have been applied in resilience and urban governance, few studies have explored the integration of LLMs with knowledge graphs to assist in devising resilience enhancement strategies. In particular, within the specialized domain of automated knowledge graph construction, there remains a lack of comprehensive frameworks for the deep integration of LLMs. While considerable attention has been paid to the general capabilities of LLMs, their potential in the precise and critical field of automated knowledge graph construction has been largely overlooked.

3. Methodology

The knowledge graph is applied to reflect BHR response process and understand the intrinsic and interactive relationships in the BHR ontology. This study proposes an automated knowledge graph construction method empowered by LLMs, and the construction steps are shown in Fig. 1.

3.1. Crawling and preprocessing of multi-modal unstructured data

Structured data can enhance the efficiency of building knowledge graphs because it often inherently includes ontologies and entities [55]. However, in most cases, data exists in an unstructured form [56], especially in emerging fields where structured data may not be readily available. Hence, extracting information from unstructured data to construct knowledge graphs has become an important area of research. Unstructured data, such as news and reports, reflects significant societal issues and serves as not only an important source of knowledge provided by the public but also a valuable channel for information acquisition [57]. Due to the emerging nature of the BHR field examined in this study, there is a lack of rich, structured data to support it. Therefore, this study focuses on collecting unstructured data related to BHR to support the construction of the knowledge graph. This study utilizes Python and APIFY [58] to crawl multi-modal data, including news, images, and videos, as our data sources from social media for constructing knowledge graphs.

For the news text data, APIFY was directly used to crawl the main body of the news, and preprocessing tasks such as deduplication, denoising, and screening were carried out on it. For the image data, a crawler was used to extract the first image of the news, and ChatGPT-4o was employed to generate descriptive information based on the features and themes of the image, thus converting the information contained in the image into text. For the video data, the media API was used to obtain all the news within the screening time range and conduct manual screening. Then, OCR (Optical Character Recognition) and speech recognition tools were utilized to extract the subtitle information, transforming the video data into text. The process of obtaining and processing the multimodal data in this paper is shown in Fig. 2.

3.2. Ontology design

The ontology is the foundation for constructing a knowledge graph, as it defines the standards for knowledge sharing within a specific field

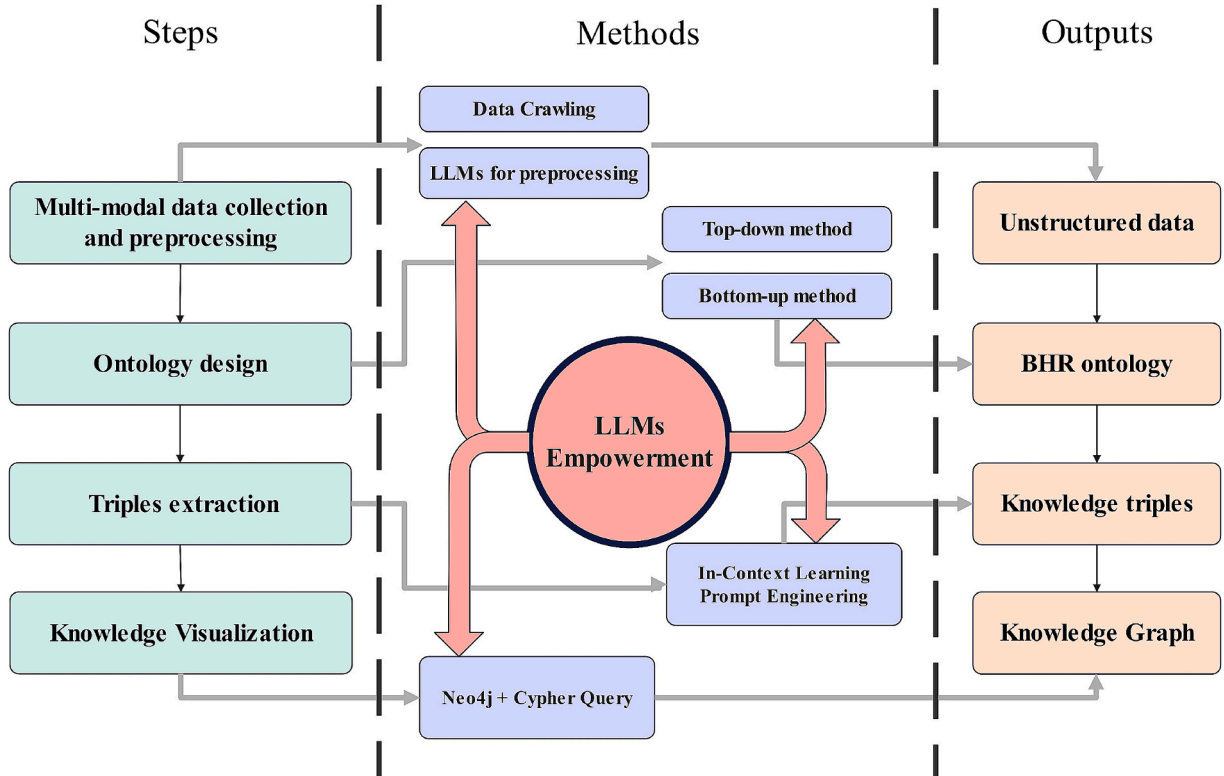


Fig. 1. BHR knowledge graph construction steps.

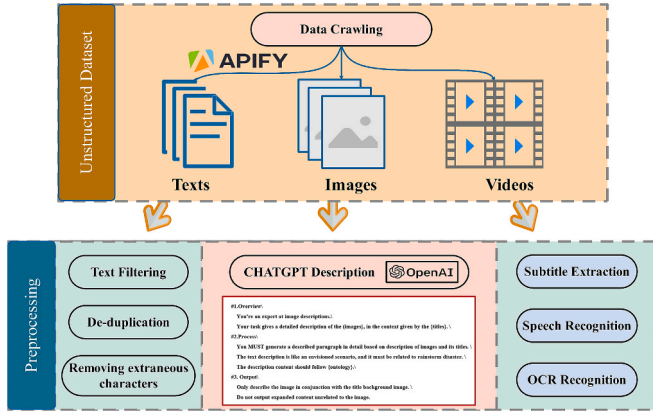


Fig. 2. Data collection and preprocessing.

[59]. It encompasses the field's fundamental concepts and their collections, as well as the definitions of concept relations and extension rules. This study employs a combination of top-down and bottom-up methods for BHR ontology design. The principle for construction is to propose the top-level design of the ontology based on the knowledge concept of BHR (Top-down) and then assist in the construction of the hierarchical structure of the ontology by building a corpus with existing data (Bottom-up). This approach combines theoretical underpinnings with data-driven techniques, ensuring the ontology is both grounded in expert knowledge and sensitive to the specific nuances within the corpus. It enhances the accuracy and comprehensiveness of the knowledge graph, reinforcing its utility as a robust tool to analyze and address complex issues within the BHR field.

$$O_{BHR} = \{KG_{concept}, KG_{relations}, KG_{property}, KG_{entity}\}$$

While $KG_{concept}$ represents the collection of all relevant knowledge concepts under the proposed topic, such as the disturbance of a rainstorm in BHR, including the definition of BHR ontology knowledge and other extracted topics; $KG_{relations}$ represents the semantic relationships between BHR knowledge under the disturbance of a rainstorm, such as hierarchical relationships and parallel relationships; $KG_{property}$ describes the attributes of BHR knowledge under the disturbance of a rainstorm, usually represented as text strings and numbers, as well as media images; KG_{entity} represents a specific instance of BHR knowledge under the disturbance of a rainstorm, such as black rain, building collapse, etc.

3.2.1. Top-down method

The top-down approach begins with an analysis of the concept of BHR. At the “building health” level, the health of the building and its occupants is considered a holistic entity, referred to as the building system in this study. “Resilience” occurs only when the building system is faced with disturbances and threats, during which the health state of the building system will change.

The entire process of BHR mainly consists of three stages. Firstly, there is the preparation stage: the health status of the building system can be automatically maintained and improved, remaining at a normal level and being unaffected by external disturbances and threats. The resistance stage refers to the situation where, when the building system faces external disturbances, its health status is affected and reaches the lowest point of its health status at the last moment of this stage. In this stage, the disaster-inducing factors and the affected objects are the key focuses. The recovery stage is when the health status of the building system begins to improve until it is fully restored, and the external threat disturbances gradually disappear. In this stage, emergency response measures and other measures (such as post-disaster repair measures) need to be taken to restore the health status of the building to the normal level.

The entire BHR cycle has similarities with the risk disaster ontology

[40]. To describe the BHR ontology through a hierarchical structure, according to the connotation of the BHR life cycle, the upper framework of BHR should include the following parts: internal or external disturbances to the building, changes in the health status of the building and its occupants, factors influencing the change in health status of inhabitants, and measures taken to cope with disturbances to restore the normal functions of the building.

3.2.2. Bottom-up method

This study applies topic modeling techniques to categorize the conceptual characteristics of BHR and lists the important terminology in BHR to assist in the tiered construction of the BHR ontology. Ontology construction requires a strong semantic expressive capacity, capable of eliminating semantic differences between concepts at different levels [59]. In addition to the top-down BHR ontology construction based on concept analysis, it's necessary to extract the various concepts represented in the BHR field knowledge to assist in the decomposition of the ontology's hierarchical structure. Because BHR is an emerging field, no scholar has yet structured the decomposition of BHR's knowledge ontology.

In order to accurately extract BHR-related concepts from unstructured text, this study employs the ChatGPT-empowered BERTopic [27] for knowledge extraction. The BERTopic model [60] is developed based on Top2Vec, utilizing a word embedding-based topic word extraction algorithm. This algorithm integrates the latest NLP techniques and, compared to traditional topic models like LDA (Latent Dirichlet Allocation), it plays a more effective role in mining the deep semantic representations of texts, enabling the extraction of more representative topic words. As a representative product of LLMs, ChatGPT excels at numerous NLP tasks, such as translation, writing, etc. Although ChatGPT's overall development is not mature, with a tendency toward “hallucination” in its outputs, it has already demonstrated excellent performance in text summarization [61]. Combining ChatGPT with BERTopic can address some of these limitations. For instance, the BERTopic model requires high-quality input texts and has limited capability in analyzing overly lengthy texts. However, a ChatGPT-empowered BERTopic model can extract topic words more comprehensively and accurately. The steps of the ChatGPT-empowered BERTopic model are shown in Fig. 3.

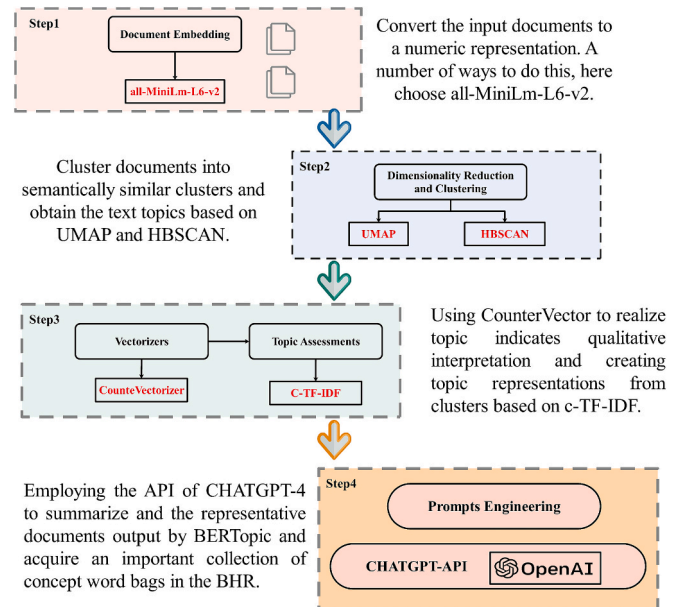


Fig. 3. ChatGPT-empowered BERTopic model framework.

- (1). *Document Embedding*. Input textual data is embedded into a language that a computer can understand. To ensure the embedding vectors retain the original semantic features of the text as much as possible, the “all-MiniLM-L6-v2” model pre-trained by Microsoft is used. This model is a pre-trained large language model, excels in semantic understanding, making it suitable for processing multimodal data, which enables it to better capture the semantic relationships among concepts related to BHR.
- (2). *Dimensionality Reduction and Clustering*. Generally, the dimensionality of text data after embedding is high. BERTopic employs the “UMAP” algorithm to reduce the dimensionality of vectors. Its superiority in preserving data structure and distribution allows for clearer visualization of high-dimensional datasets, facilitating subsequent analysis. The “HDBSCAN” algorithm was chosen for clustering, discovering potential topics through textual vector features. Due to its low sensitivity to noise, it can effectively identify complex cluster structures in real-world data, thereby improving clustering accuracy [62].
- (3). *Tokenization and Topic Assessment*. The “CounterVectorizer” was used because of its simplicity and high efficiency, its ability to perform word frequency statistics, its excellent compatibility with other models, and its strong customizability. It can quickly tokenize the topics clustered by BERTopic and assist in the extraction of key information and subsequent analysis during the construction of the knowledge graph of BHR. “c-TF-IDF” was used (a variant of TF-IDF) to evaluate the importance of each word within each HDBSCAN cluster, selecting representative words for each topic for thematic representation. This algorithm emphasizes keywords with high frequencies within specific classes rather than across all documents, aligning well with the feature extraction requirements of our knowledge graph centered on building health resilience.

$$C - TF - IDF = \frac{TF}{\sum(TF)} * \log\left(\frac{N+1}{N_c+1}\right)$$

Where TF (Term Frequency) represents the frequency of a term within a 360 document, $\sum(TF)$ represents the sum of term frequencies within the document, N represents the total number of documents, and N_c represents the number of documents within the cluster or class.

- (4). *Topic Extraction*. By inputting topics and representative documents, prompts are constructed, and the ChatGPT-4o API is utilized to summarize and synthesize the documents, resulting in the final compilation of a topic word library.

Finally, the top-level concepts of BHR are analyzed using a top-down approach, while the bottom-up approach is employed to hierarchically decompose BHR knowledge. The results of these two methods are then integrated and refined through mutual supplementation and matching of concepts and subcategories, culminating in the final BHR ontology.

3.3. LLMs for entity recognition and relation extraction

After constructing the ontology, it is necessary to identify the entities corresponding to the concepts as well as the relations between these entities. However, in an emerging field, extracting entities and relationships from unstructured text using traditional identification methods is complex and requires extensive manual annotation. LLMs exhibit good performance in text generation and question-answering tasks. There are many LLMs in today's market, among them, ChatGPT is a popular AI tool proposed by OpenAI [63]. ChatGPT operates on a generative artificial intelligence-based question-and-answer system, and it possesses significant abilities in translation, creating stories, debugging code, and other aspects [64]. Particularly regarding summarization capabilities based on text, many scholars have demonstrated that

ChatGPT performs well in analyzing and understanding natural language and summarizing the main points according to semantic relations [61,65].

Constructing a knowledge graph from unstructured text is exceedingly complex, requiring extensive manual annotation of entities and relationships. Even semi-automatic supervised algorithms like BiLSTM-CRF and BERT-BiLSTM-CRF for NER need certain annotated data. This process can consume considerable human effort. Despite some “hallucination”, ChatGPT-4o, as a generative AI tool, is highly accredited for its capacity to extract knowledge from unstructured text automatically [66]. ChatGPT-4o excels in text summarization and semantic analysis due to its advanced neural architecture, which enables deeper contextual comprehension, improved coherence, and more accurate extraction of key information. Its ability to understand nuances, identify relationships between concepts, and generate concise yet comprehensive summaries makes it highly effective for tasks requiring precise language processing [67]. Additionally, its enhanced training data and optimization techniques contribute to better generalization across diverse topics, ensuring more relevant and insightful responses. Using ChatGPT-4o for named entity and relationship recognition not only greatly saves on human annotation efforts but also better understands the contextual semantic relationships in any field's unstructured text. It is not constrained by a specific set of concepts, which is an effective ability that pre-trained NLP models cannot achieve [68].

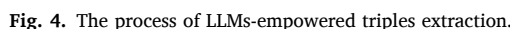
Utilizing the ChatGPT-4o application programming interface (API) to summarize unstructured knowledge within unstructured texts allows for the refinement and structuring of this knowledge. This article leverages ChatGPT-4o to propose a model for entity and relationship extraction based on feedback checking. The specific content of the model is shown in Fig. 4.

Store the processed multi-modal data in text form and employ prompt engineering to enable ChatGPT to automatically extract triples from the text data. For each text document, ChatGPT follows these steps:

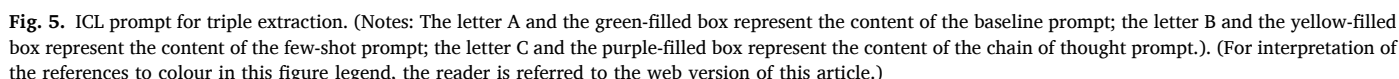
- (1) *Triple Extraction Prompt Engineering*: 1) Ontology framework learning: Start by inputting the ontology framework. This ensures that the recognized entities and relationships fall within the definitions provided by the ontology. 2) Annotation case learning. Using three real news articles as examples, manually annotate all triples. These examples will serve as cases for ChatGPT to learn from. 3) Automated Triples Extraction: ChatGPT automatically extracts triples using the developed prompt engineering strategy. The triples should then be saved in JSON format.
- (2) *FCM Prompt Engineering*: Use the initial triple extraction results alongside the original document as inputs to construct prompts. 1) Triples supplement: This helps in reprocessing the triples. Begin by comparing the initial triple results with the original document to check for any omissions. 2) Semantic consistency checking. Verify the relationships between entities and concepts using semantic checks to ensure accuracy. 3) Deletion of Redundancy. Delete any triples that have the same overall meaning to eliminate redundancy. 4) Similar Entities Merging. Further, filter the extracted entities by checking their semantic similarity and consistency, and merge the entities with the same semantics. By following these steps, the process ensures that the extraction and refinement of triples are both systematic and accurate, resulting in a well-structured and reliable triple data set.

3.4. In-context learning prompts

To enhance LLMs' reasoning abilities of entities and relations extraction for BHR, especially, this study employs ICL-based prompts to train the LLMs. ICL has been popularized as a method for learning tasks using LLMs in GPT-3. Essentially, it uses a pre-trained language model to conditionally generate probabilistic distributions based on given



refining the underlying logic for inferring entity relationship triples of BHR.



- (1) **Baseline Prompt.** Fig. 5A illustrates the framework for the baseline prompt, which primarily includes the task objectives for CAHTGPT-4o, components for context-aware relationship extraction such as triples formed by nodes, node types, and edges, as well as a set of constraints for the extraction process. Additionally, the output format is also defined.
- (2) **Few-shot Prompt.** Fig. 5B shows the content of the few-shot prompt. This section mainly provides the detailed extraction process and also examples of entities and relations extraction to help the LLM better understand the task.
- (3) **Chain-of-Thoughts Prompt.** Research has demonstrated that the chain of thought can assist LLM in generating complex and logically consistent responses, thereby improving the accuracy of task outputs [50]. The reasoning steps for the chain of thought are presented in Fig. 5C.

4. Results

4.1. Case study

The automated knowledge graph construction method for BHR is applied to generate BHR knowledge graph under rainstorms in Hong Kong, validating the effectiveness of the proposed method. Compared with the domain-specific documents that are difficult to obtain, news data can more truly reflect the health status of buildings and the health status of building occupants when disasters happen. The texts of the news related to building health resilience on the HK01 website during this period, the images of the news, and all the news videos from the official HK01 YouTube account were collected. The samples of the collected multimodal data are shown in Fig. 6. The reasons why adopting rainstorms of Hong Kong as the study case are as follows: 1) Diverse and dense urban environment. Hong Kong is a high-density megacity in Asia with a variety of building types, such as public housing, high-rise buildings, low-rise buildings, and villas. The high density and complexity of urban buildings make BHR more crucial for urban governance in Hong Kong. 2) Frequent rainstorm disasters. Due to the

interaction of maritime and land influences in lower latitudes and the South China Sea monsoon, Hong Kong experiences frequent rainstorm disasters [70], especially during September 2023, when Hong Kong has long suffered from Typhoons and centennial rainstorms (<https://www.hk01.com>); 3) Rich and representative data source. Due to the scarcity of research related to BHR, it is not feasible to effectively obtain structured knowledge from literature and structured databases. HK01 is one of the most mainstream media outlets in Hong Kong, providing rich and representative information [57]. In addition to text, media data includes valuable knowledge in the form of headline images and news report videos. This knowledge can compensate for the limitations of text alone, making the data sources for constructing knowledge graphs more diverse.

By using these data sources and methodologies, this study aims to build a comprehensive knowledge graph to enhance the understanding and management of BHR. This study has crawled a total of 314 news text articles related to rainstorms from the HK01 website from September 2023, 312 news images, and 52 related videos from the HK01 YouTube channel. The data was preprocessed according to the steps outlined in Section 3.1.

4.2. BHR ontology under rainstorms in Hong Kong

This study takes the rainstorm in Hong Kong as an example of disturbance and constructs the ontology of BHR using the top-down approach described in section 3.2. Fig. 7 shows the changes in the health status of buildings under rainstorms. The entire process of BHR resilience is divided into three stages. In the preparation stage, the rainstorm has not yet occurred but is in the process of forming. At this time, both the natural and human environments become factors that incubate rainstorms. In addition to natural factors, human activities leading to climate change can also indirectly increase the frequency of rainstorm disasters. The second stage is the resistance stage, in which the rainstorm has already occurred, leading to the destruction of building infrastructure, urban waterlogging, landslides, and other risks that threaten the health status of occupants and buildings (such as injuries or



Fig. 6. HK01 collecting data samples.

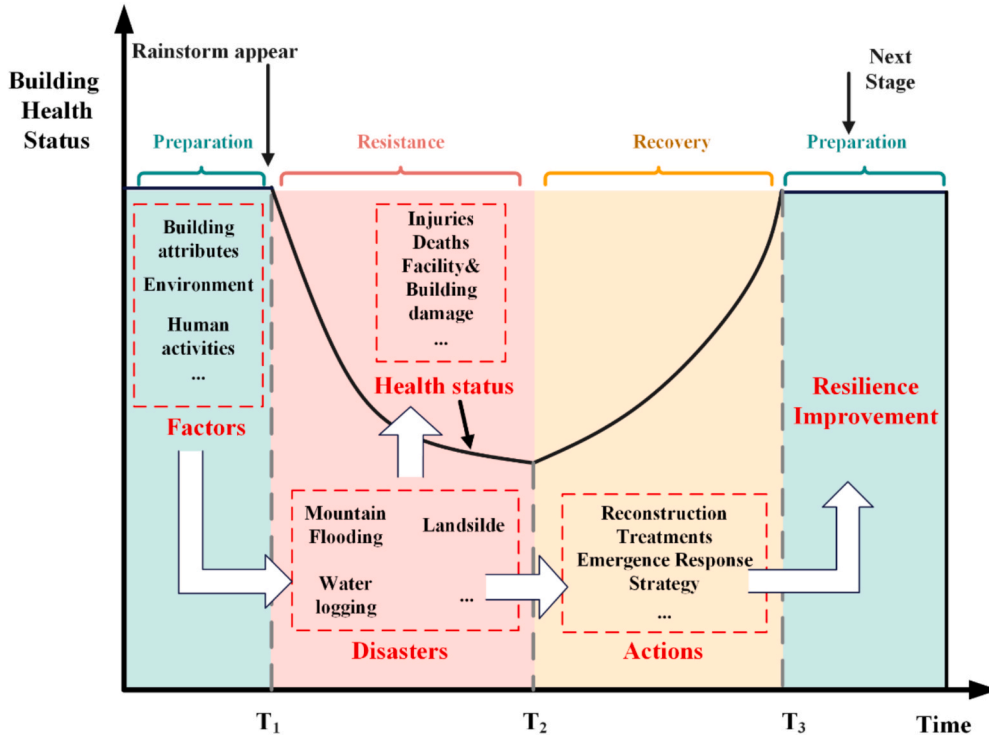


Fig. 7. The process of BHR under rainstorms, adapted from (Zhang et al., 2024).

deaths of residents, building collapses, etc.). The third stage is the recovery stage, where all health status of the building system is restored by itself. It is also necessary to take some emergency measures, post-disaster remedial measures, and develop preventive measures for facing the next disaster to improve BHR. In summary, the entire cycle of BHR under rainstorms includes the four most important $KG_{concept}$ in the top class: Disasters, Factors, Health status, and Actions.

Section 3.5 introduces the ChatGPT-empowered BERTopic model, which extracts topic words from unstructured text to assist in bottom-up ontology construction. Post-processing of unstructured text data and input into the BERTopic model reveals results as shown in Fig. 8. Within the configuration of the BERTopic model, “min_topic_size = 10” allows for the extraction of twenty relevant topic words. Fig. 8A displays the similarity among various topics, notably *Topic1(warning, observatory, rainstorm)* and *Topic12(rain observatory, heavy)*, which exhibit the highest degree of similarity, reaching 0.88. Fig. 8B presents the most representative words of the top eight topics, offering insights into the essence of each topic. For instance, *Topic 0* is related to government operations, while *Topic 2(road, car, water)* and *Topic 11(mtr, station, wong)* are related to transportation infrastructure, focusing more on roads and subways, respectively. Fig. 8C illustrates the distribution and clustering of representative documents for each topic. These documents provide a categorical basis for constructing the hierarchical structure of BHR ontology under the impact of rainstorms.

This study applies the bottom-up method to assist the segmentation, which involves extracting related concepts from the collected unstructured data. Using the topics and representative documents output by BERTopic and applying LLMs to summarize the content, the topics are extracted from this process, illustrated in section 3.2.2. Integrating these topics with the concepts in Fig. 7, the ontology hierarchy $KG_{concept}$ under “rainstorm” is obtained as shown in Table 2.

The entities are included in every concept of the ontology and there are also many relations between entities. Fig. 9 illustrates the addition of entity objects and relationships to the constructed ontology as an example based on the constructed ontology.

This example primarily describes the disasters such as “black rain”

and “typhoons” that occur when the building systems in Hong Kong face the disaster of a “rainstorm”. Except for the $KG_{concept}$ shown in Table 2, the other squares represent the identified entities KG_{entity} , and the connecting lines in the diagram represent the relations $KG_{relations}$. Under the disturbance of such disasters, the structural integrity of buildings can be compromised. Additionally, this could lead to “class suspension”, “occupants catching fever”, and other situations, adversely affecting the health status of the building occupants. The change in health status is due to the disturbance caused by the rainstorm, which leads to drastic changes in the living environment and climate. The figure also shows actions for minimizing the disturbance to the health status of buildings and occupants as much as possible, measures such as more “accurate and timely weather warnings” and “emergency evacuation” after the occurrence of a rainstorm can enhance the resilience of the health status of buildings and occupants. The ontology mentioned above can comprehensively describe the entire process of resilience changes in building health under the disturbance of a rainstorm and various entity concepts, guiding the construction of knowledge graphs.

4.3. BHR knowledge graph under rainstorms in Hong Kong

According to the guidelines established in Sections 3.3 and 3.4, this study extracted 4,163 BHR triples under rainstorms in Hong Kong. This large number of triples contains not only a vast amount of entity knowledge but also complex relational features. To visualize the knowledge graph of BHR, this study utilizes Neo4j to store the entities and relation of the triples. Neo4j uses a topological graph structure to store knowledge on triples, allowing for an intuitive representation of entities and their interrelations [39].

By conducting a statistical analysis on all entities obtained from the triples under the top class “disasters”, it was found that “Floods” and “Landslides” are two frequently occurring secondary disasters in Hong Kong under rainstorms. Fig. 10 shows the knowledge graph queried using Cypher query related to these two disasters. The nodes of different colors in the graph represent different ontology types. The knowledge graph includes the causes of floods and landslides disasters (light yellow

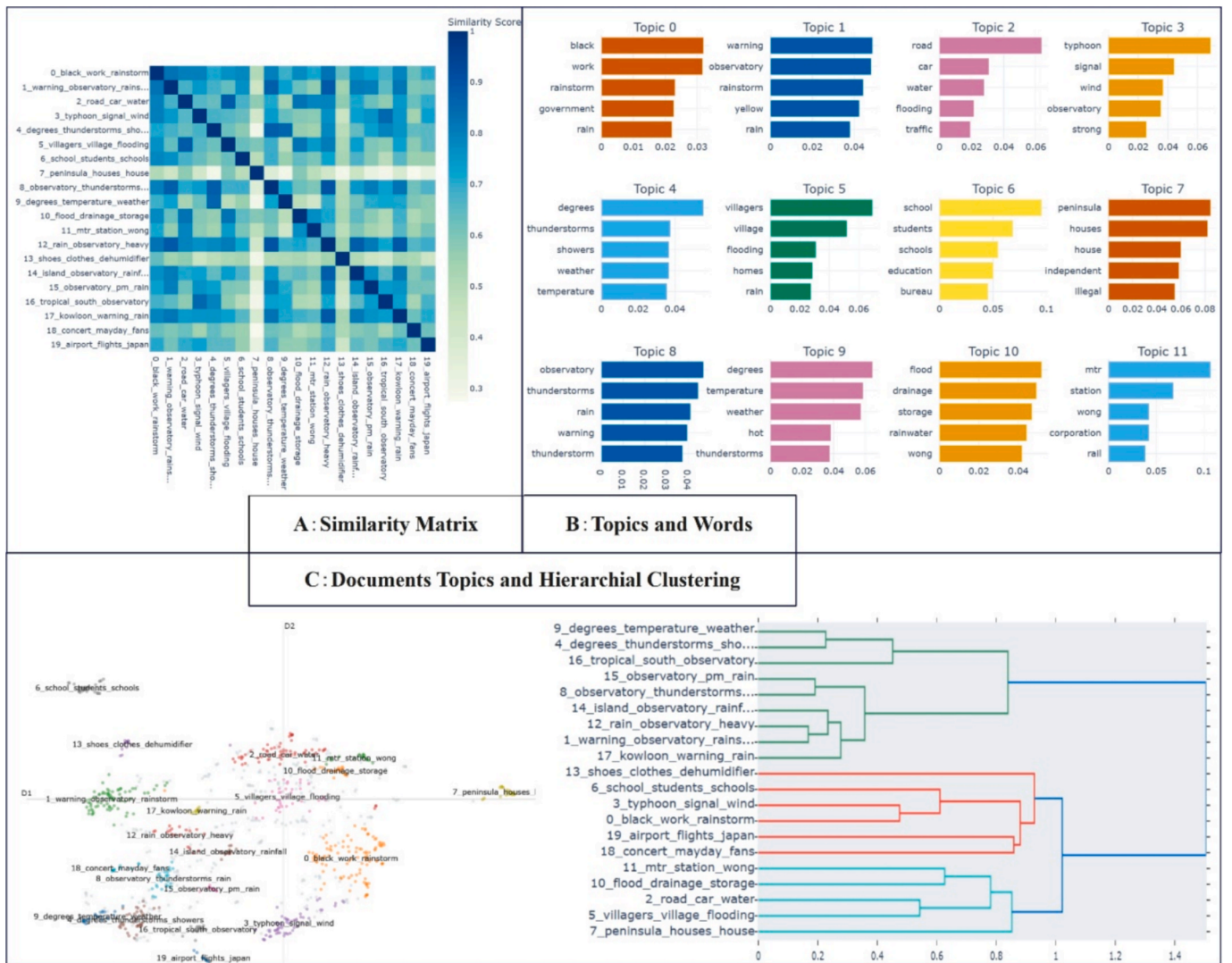


Fig. 8. The result of BERTopic. (Notes: This figure demonstrates three visual outcomes of the BERTopic model. In A, all 20 extracted topics are displayed along with their pairwise similarity degrees, where darker squares indicate higher similarity. B showcases the top 12 topics along with the top 5 representative words for each topic, ranked in order of significance. C presents the results of topic clustering, categorizing the topics into three main groups, with the distribution of relationships among the topics also illustrated.).

nodes), the impact on building health status (green nodes), and measures to deal with disasters (blue nodes). The detailed knowledge representation of BHR related to “floods” and “landslides” is shown in Fig. 11.

Fig. 11 takes the influencing “factors” of floods, “health status”, and the affected “locations” of landslides and their impact on “health status” as examples to illustrate the knowledge of BHR. In the “FLOODS-Locations” subgraph, specific addresses in Hong Kong affected by floods are displayed. This knowledge subgraph reveals that floods have impacted areas such as “Shek O” and “Wong Tai Sin” in Hong Kong. In the “FLOODS-Health status” subgraph, the health status of occupants, buildings, and the surrounding environment affected by floods is presented. For buildings, floods can cause physical disruptions such as “Destruction of buildings” and “Cracked walls and broken windows” which, in turn, affect the health of the surrounding environment, leading to issues like “Garbage scattered” and “Closure of stores”. For the occupants within the buildings, floods also impact their health, causing conditions such as “Occupant anxiety” and “Occupant tension and confusion”. In the LANDSLIDES-Factors subgraph, it can be observed that, in addition to rainstorms, there are other climatic causes for landslides, such as “Extreme weather”, “Typhoons” and “Black rain”.

Besides climatic reasons, factors like “Pipe drainage”, “Unauthorized buildings”, and “Geologically vulnerable” are also significant drivers of landslides. In the LANDSLIDES-Health status subgraph, the impact on the health status of buildings, the environment, and occupants due to landslides is similar to that caused by floods. The physical health impacts on buildings include “Building shaking” and “Exposed and overhanging foundations”. For infrastructure, the impacts include “Traffic disruption”, “Network outage”, “Blackouts”, and “Stopped drinking water supply”.

The above subgraphs are part of the BHR knowledge graph under rainstorms from HK01 multi-modal data, providing formation on factors, affected locations, and impact results of disasters. Therefore, the knowledge graph is of significant reference value for understanding BHR under rainstorms. It allows the analysis of structured knowledge to take appropriate governance measures.

5. Discussion

5.1. Validation of triples extraction

Traditional methods for constructing knowledge graphs often rely on

Table 2The ontology hierarchy of BHR *KG_{concept}* under a rainstorm.

Top_class	Sub_class	Description
Disasters	Black Rainstorm Warnings	Occurrences of black rainstorm warnings and their impact on various areas, including disruptions to trading and public services.
	Typhoon Haima and Hato	Impact of Typhoon Haima and Hato on Hong Kong, causing severe weather conditions and flooding.
	Flooding and Emergency Measures	Reports of flooding in various areas and the emergency measures deployed, including the opening of temporary shelters.
	Impact on Transportation	Effects of rainstorms on transportation, including MTR disruptions, road blockages, and impacts on air travel.
Health status	Health and Safety Advisories	Health and safety advisories are issued by the government during rainstorms, including warnings to stay indoors and updates on public health services.
	Economic Impact of Rainstorms	Economic consequences of rainstorms include trading delays and the financial impact on businesses.
	Community and Social Impact	The social impact of severe rainstorms on communities, including displacement and the need for temporary shelters.
	Infrastructure and Building Safety	Concerns and measures related to the safety of buildings and infrastructure during severe weather conditions.
	Educational Disruptions	Impact of rainstorms on educational institutions, including class suspensions and safety measures for students.
	Community Resilience and Adaptation	How communities adapt to and recover from the impacts of severe rainstorms, including community resilience strategies.
Factors	Rainstorm Warning Systems	The role and effectiveness of rainstorm warning systems in Hong Kong, including different levels of warnings issued by the Observatory.
	Weather Predictions and Alerts	The role of weather predictions and alerts in preparing for and responding to severe weather conditions.
	Environmental and Climatic Factors	Discussion on how environmental and climatic factors contribute to the severity and frequency of rainstorms in Hong Kong.
	Long-term Weather Impact Studies	Studies and research on the long-term impacts of severe weather on Hong Kong's environment and society.
Actions	Public Awareness and Preparedness	Efforts to raise public awareness and preparedness for dealing with severe rainstorms, including educational campaigns.
	Legal and Policy Framework	Legal and policy frameworks are in place to manage and mitigate the effects of rainstorms and related disasters.
	Technological Solutions	Use of technology in monitoring, predicting, and responding to severe weather conditions.
	Emergency Services and Response	The role and effectiveness of emergency services during rainstorms, including rescue operations and coordination.
	Government Response to Rainstorms	Actions taken by the government and public departments in response to severe rainstorms, including trading suspensions and public service adjustments.
	Public and Employee Safety	Guidelines and measures taken to ensure the safety of the public and employees during rainstorms, including work suspension advisories.

manual annotation and supervised deep learning techniques to extract knowledge entities and relationships. In this study, the construction of the knowledge graph primarily relies on triples extracted from multi-modal data using the LLMs API. Therefore, it is necessary to verify the effectiveness of the proposed method for constructing the knowledge

graph in this study. Typically, the most common approach is to compare the results extracted by ChatGPT-4o with those extracted manually. However, the aim of this study is realizing automated knowledge graph construction, which means the evaluation process also need to be automated [71]. Therefore, this study constructs an automated validation of triple extraction accuracy from coverage and consistency perspectives through three approaches: Manual Evaluation, Metrics Evaluation and LLMs-based Evaluation.

5.1.1. Manual evaluation

This study systematically validated the performance of ICL&FCM-driven framework for automatic triples extraction from text documents using LLMs. To ensure robustness, a stratified random sampling approach was employed to select 10 representative documents for manual annotation. Human annotators, trained in entity recognition and relation extraction, independently extracted triples from the selected texts. Quantitative comparison between LLM-generated triples and human-annotated shows that there is a high degree of similarity between the triples extracted manually and those extracted by the LLMs model, indicating the relatively high reliability of the LLMs in extracting triples. However, some differences also exist.

The discrepancies observed were primarily attributed to ambiguous entity references (e.g., pronouns) and context-dependent relations (e.g., temporal or causal connections), which were mitigated by the FCM's contextual reasoning module. These results align with recent advancements in few-shot information extraction [72]. The validation framework underscores the efficacy of integrating ICL and FCM for triples extraction, providing a replicable methodology for future studies.

5.1.2. Metrics evaluation

This study utilizes metrics from the NLP text generation [73], including BLEU [74], ROUGE [75], METEOR [76], BERTScore [77]. BLEU and ROUGE metrics are based on the n-gram overlap between the input and output contents of ChatGPT-4o. BLEU is commonly used for translation evaluation, while ROUGE is used for summarization evaluation; METEOR and BERTScore are similar. METEOR evaluates translation quality by taking into account synonyms and syntactic structures, going beyond simple lexical matching. BERTScore is based on a pre-trained deep learning model that better captures semantic similarity, thereby evaluating text similarity.

Table 3 illustrates the scores of NLP-based metrics for extracting triples from multi-modal data (text, image, video). The variables “Text_triples” and “Text_triples_FCM” denote whether feedback check mechanisms were incorporated during the triple extraction process. From the results, it can be observed that compared to text and image data, the BLEU and METEOR scores for extracting triples from video data are the highest. Specifically, incorporating FCM increased the BLEU score to 11.02 % and the METEOR score to 19.28 %. However, the performance of extracting triples from video data in ROUGE is noticeably lower than that of other modalities. This indicates that while the triples extracted from video data have higher coverage, they are less effective in summarizing the original text. Furthermore, the inclusion of FCM resulted in an overall improvement in the metric scores of triples extracted from all three data types. For instance, the ROUGE-1 score for extracting triples from text data increased by 2.8 % after incorporating FCM. In the BERTscore metric, although there was a slight decrease in Precision for all data, Recall and F1 scores improved with the inclusion of FCM. This suggests that integrating FCM during the LLMs-based triple extraction process is essential for enhancing both precision and coverage.

5.1.3. LLMs-based evaluation

Another unsupervised and automatic evaluation method is called LLMs-based Evaluation (LLMs-Eval) which uses LLMs to evaluate the result of LLMs. ChatGPT-4o is employed for LLMs-Eval. Due to the high cost of manual annotation for context-aware knowledge graphs,

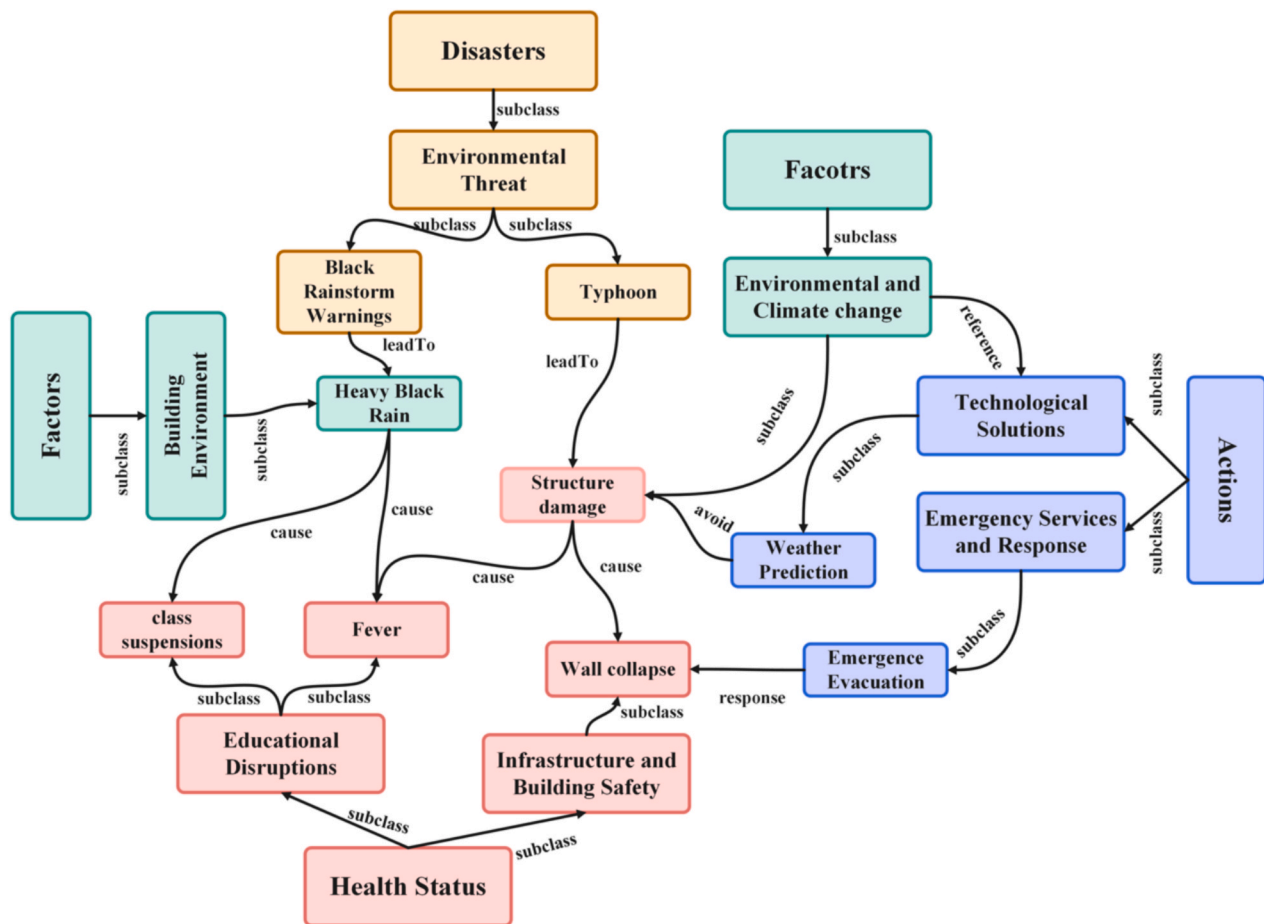


Fig. 9. The example of entities and relations in BHR ontology. (Notes: The figure illustrates instances and relations of four ontologies within BHR. This case demonstrates that the constructed BHR ontology covers the entire response process of BHR under the disturbance of a rainstorm. Pink rectangles represent the “Health Status” concept and the entities it includes, green represents the concept and entities related to “Factors,” brown represents concepts and entities associated with “Disaster,” and blue-violet represents the concept and entities related to “Actions”). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

leveraging GPT for evaluation is more effective. Existing studies have shown that GPT performs well in evaluating generation text from consistency, fluency, and coherence aspects [78]. Furthermore, Wu has demonstrated that LLMs-Eval assessments exhibit a high degree of consistency with manual evaluations [71].

Therefore, this study also employs LLMs-Eval for the evaluation of triple extraction. Compared to zero-shot and few-shot prompts, the ICL prompting engineering yields better results. This study designs ChatGPT-4o to evaluate the quality of LLMs outputs and give a score from 1 to 5 following two aspects: 1) Evaluating the extracted triples from the perspective of coverage to verify whether the extracted content covers the knowledge of the original text; 2) Assessing the consistency of the triple entities to ensure they correspond with the concepts in the ontology. The ICL prompts designed based on the above criteria in [section 3.4](#) are shown in [Fig. 12](#).

As shown in Table 4, among the three kinds of modal data—text, image, and video—the highest LLMs-Eval score is for image data. This differs from the results of NLP metrics, where video data scored the highest. The high proportion of video data receiving a score of 1 in LLMs-Eval may be due to the smaller data volume. In this study, only 52 video samples were collected, which is significantly fewer compared to the number of image and text samples. However, with equal data volumes, the precision of extracted triples from image data is higher than that from text data, as consistently indicated by both LLMs-Eval and NLP metrics. Additionally, the average LLMs-Eval scores for image and video data show a slight improvement after incorporating FCM. Although the

average score for text data decreased, the proportion of scores of 1 reduced, and the proportion of scores of 5 increased by 1 %. Therefore, overall, FCM contributes to improving the performance of triple extraction.

5.2. Organizations and actions in BHR knowledge graph

The organizations are the main subject of the implementation of the strategies. The most crucial aspect of studying strategies to enhance BHR under rainstorms is learning from historical experiences. Through the knowledge graph, past countermeasures can be structurally displayed. This study utilizes Cypher queries to identify specific actions and the implementing organizations aimed at improving BHR under rainstorms. The query results are shown in [Fig. 13](#).

Fig. 13 illustrates that multiple organizations in Hong Kong have participated in enhancing BHR, including the “Government”, the “Observatory”, the “Education Bureau”, the “Buildings Department”, and various other management institutions. This highlights the necessity of collaborative governance from multiple organizations when addressing rainstorms. The network connections among these collaborators can build trust, and maintaining cooperation over time can improve the efficiency of implementing disturbance response strategies [79]. Furthermore, the cooperation of multiple organizations can facilitate resource sharing and strengthen the overall rapid response capability of the collaborative network [80]. Therefore, the collaboration trend among various organizations in Hong Kong is crucial for enhancing BHR

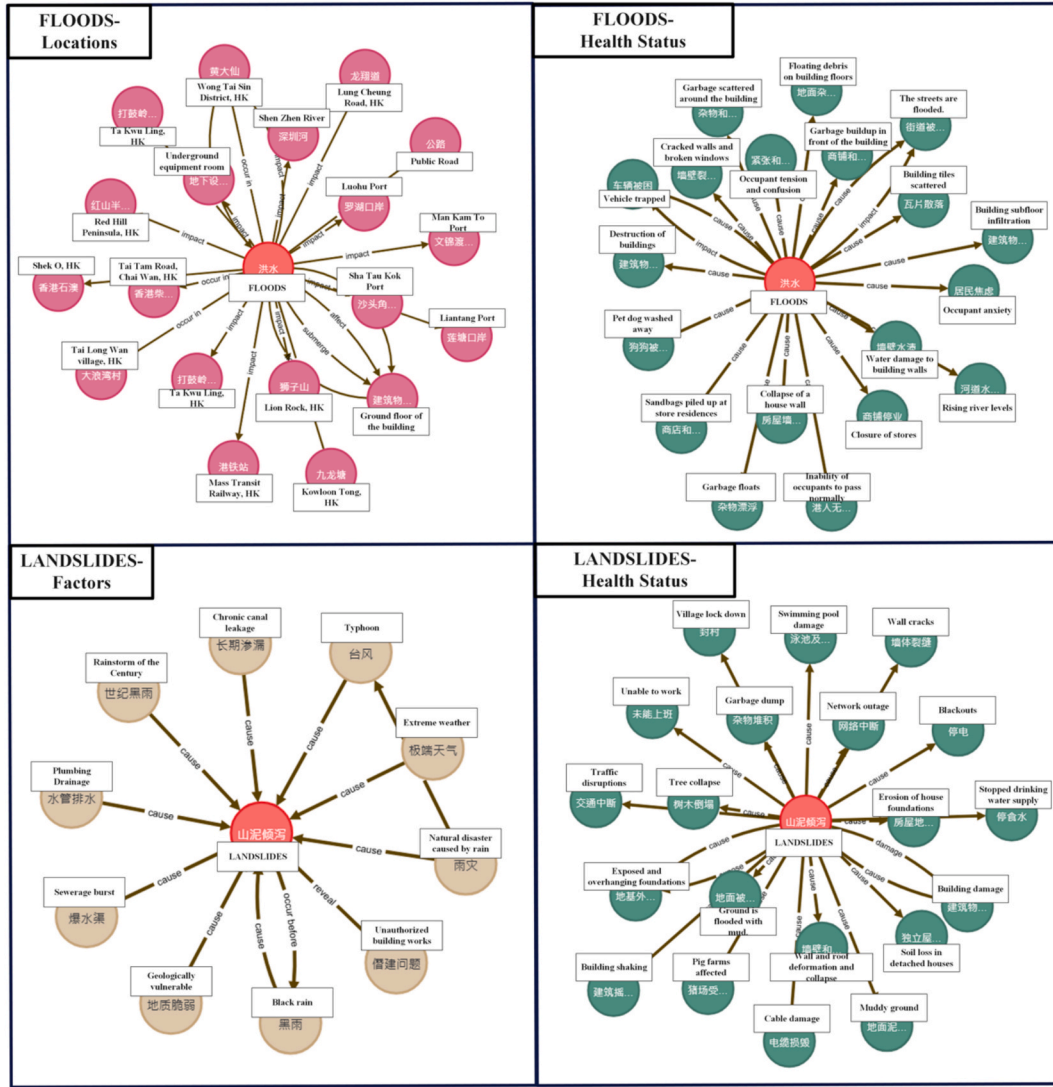


Fig. 11. Subgraphs of the knowledge graph from landslides and floods disasters.

Table 3
Result of NLP metrics for extraction evaluation.

Triples\Metrics	BLEU	ROUGE-1/L	METEOR	BERTScore			
				Precision	Recall	F1	
Text_triples	3.44	13.27	12.49	11.53	72.86	60.12	65.85
Text_triples_FCM	5.46	16.08	14.79	13.58	72.85	61.22	66.49
Image_triples	9.53	10.92	10.92	18.06	74.47	59.97	66.41
Image_triples_FCM	10.76	12.46	12.46	18.89	74.22	60.34	66.54
Video_triples	10.26	0.38	0.38	18.70	76.90	62.10	68.62
Video_triples_FCM	11.02	0.37	0.37	19.28	76.78	62.38	68.74

5.3. Implications

5.3.1. Automated construction process of knowledge graph

This study takes full advantage of emerging generative AI technology, specifically LLMs, to automatically construct the knowledge graph, thus eliminating the need for extensive manual annotation and training work. ChatGPT-4o also performs well in terms of the coverage and consistency of extracted triples. Additionally, LLMs can integrate multi-modal data, enriching the content of the knowledge graph. There are differences between multi-modal knowledge graph and knowledge graph constructed in our study. The multi-modal data is employed as data source for knowledge supplementation, rather than visualizing it in

the knowledge graph [84]. Unlike traditional research that uses computer vision to extract knowledge or features from videos and images [85,86], this study employs the ChatGPT-4o enables automated and comprehensive extraction of knowledge from multimodal data. Therefore, employing LLMs to empower knowledge graph construction is a promising research direction. It not only enhances the efficiency of the knowledge graph construction process but also enriches its content. With the continuous updates and iterations of generative AI in video and image processing, these technologies will have more innovative applications in knowledge graph construction.

Furthermore, this study employs LLM to automate the extraction of entities and relations, saving a significant amount of complex manual

#1. Overview
 You're an expert at evaluating the triples from texts.
 Your task is giving a score from [1,5] according to evaluate coverage and consistency of the triples in {graphs} extracted from {texts}.
A

#2. Evaluation Steps.
 ##1.Check that the contents of all graphs reflect the meaning of the original text, deduct points if they do not.
 ##2.Check whether the nodes of the graphs are conforming to the definition of {ontology}, if not then deduct points according to the number of non-conformities.
 ##3.Check whether the edges between the nodes of the graphs are in accordance with what is in the original text, if they are not then deduct points based on the number of non-conformities.
B

#3.. Output
 Only output the JSON format results having the following keys: "text", "graph", "score".
A

--> Beginning of Chain-of-Thought
 If the content, nodes, and edges of the graph largely conform to the explicit checking guideline above, then the graph receives a score of 5.
 Conversely if the graph has numerous content, node, or edges that conflicts with the explicit checking guideline, then the graph scores 1.
C

--> Beginning of example
#Input
 text1 = "雷雨下午影响珠江口一带,天文台曾发大雨预警.天文台今日12日下午3时发出特别天气提示,表示与低压槽相关的雷雨区正逐渐靠近珠江口一带,并可能会在未来一两小时影响香港"
 graph1 = "[('雷雨','influence','珠江口'),('雷雨','response','大雨预警'),('香港天文台','report','大雨预警')]"

text1 = "Thunderstorms are affecting the Pearl River Estuary area in the afternoon, and the observatory has issued a heavy rain warning. The observatory issued a special weather alert at 3 PM on the afternoon of the 12th, indicating that the thunderstorm area associated with a low-pressure trough is gradually approaching the Pearl River Estuary and may affect Hong Kong in the next hour or two."
 graph1 = "[('Thunderstorm', 'influence', 'Pearl River Estuary'), ('Thunderstorm', 'response', 'heavy rain warning'), ('Hong Kong Observatory', 'report', 'heavy rain warning')]"

#Output
 [{"text": text1, "graph": graph1, "score": score1}]
--> End of example
B

Fig. 12. ICL prompt for LLMs-Eval.

Table 4
Result of LLMs-Eval.

Triples/Score	LLMs-Eval Score					average
	1	2	3	4	5	
Text_triples	0.93 %	2.79 %	27.44 %	68.37 %	0.47 %	3.647
Text_triples_FCM	1.40 %	2.79 %	28.84 %	65.58 %	1.40 %	3.628
Image_triples	—	—	6.67 %	78.46 %	14.87 %	4.082
Image_triples_FCM	—	—	6.15 %	78.46 %	15.38 %	4.092
Video_triples	5.26 %	—	39.47 %	55.26 %	—	3.447
Video_triples_FCM	2.63 %	—	44.74 %	52.63 %	—	3.474

annotation work. This method is suitable for extracting triples from unstructured knowledge. Beyond triples, LLM is also available to extract other types of knowledge from unstructured texts, such as summaries, keywords, topics, etc. The structure of the knowledge graph is constrained by the ontology. While LLM can identify a rich set of triples, it cannot directly construct a knowledge graph due to its complexity. This poses higher requirements for knowledge fusion. In the process of constructing the BHR knowledge graph in this study, considerable effort has been invested in the reprocessing of triples, such as entity fusion and relations integration.

5.3.2. The application of BHR ontology

BHR is a relatively new field with interdisciplinary connections. Establishing a well-defined ontology ensures knowledge is structured systematically, reducing ambiguity and improving data interoperability across different disciplines. The construction of the BHR ontology presented in this study employs a dual-methodological framework integrating conceptual modeling analysis (Fig. 7) and topic model-based text analytics (Fig. 8). Based on this method, the ontology provides a holistic view of BHR, which is beneficial for urban planners, policymakers, and

risk management professionals in decision-making.

A case study of Hong Kong's 2023 heavy rainfall event was analyzed to validate the ontological framework. When buildings are exposed to heavy rainfall, this event is classified under the "Disasters" concept, which encompasses secondary hazards such as typhoons and flooding that propagate urban system impacts. The "Factors" category incorporates weather prediction models, environmental monitoring datasets, and peer-reviewed research findings on extreme precipitation mechanics. Given the study's focus on building resilience, the "Health Status" dimension assumes central importance, quantifying both physical structural integrity and human risk metrics. Finally, the "Actions" concept aggregates government intervention strategies, including emergency shelter deployment, public preparedness campaigns, and advanced technological solutions such as real-time rainfall forecasting systems.

This empirical demonstration illustrates how the four-dimensional BHR ontology systematically captures the multi-faceted interactions between hazard agents, contributing factors, system responses, and mitigation measures. LLMs play a transformative role in this process by enabling scalable knowledge discovery from unstructured data, thereby

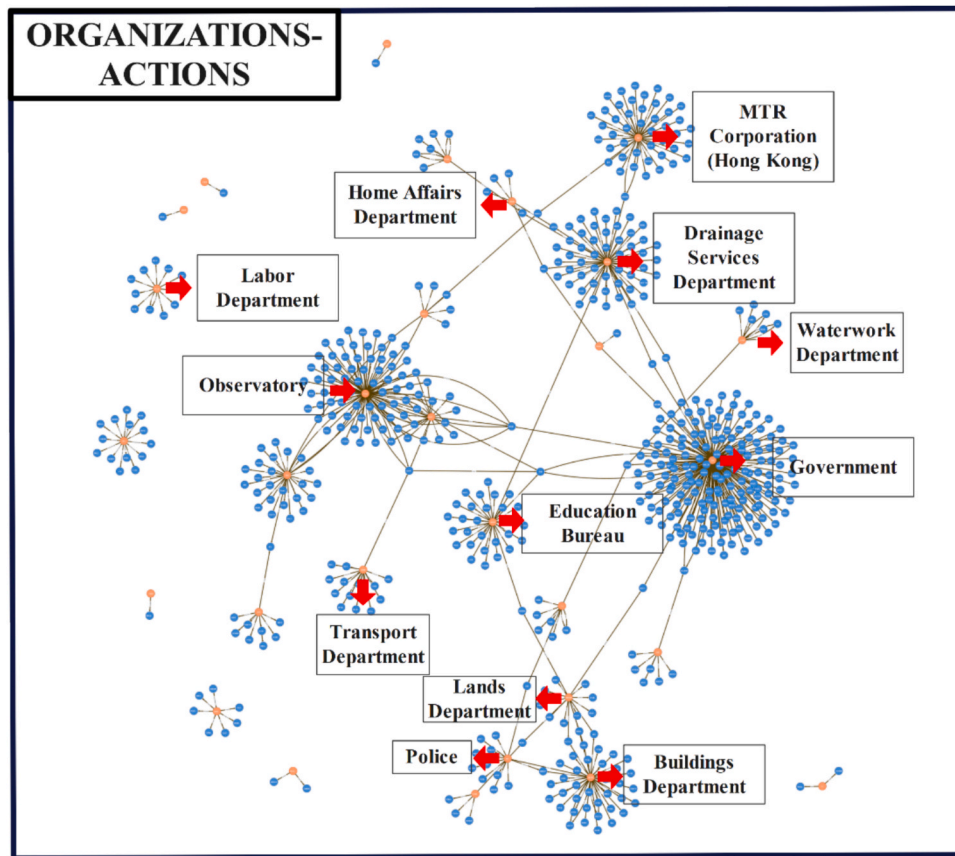


Fig. 13. Organizations and actions in BHR knowledge graph.

enhancing the efficiency and analytical depth of ontology development. This reduces the reliance on purely manual expert-driven ontology construction, enabling continuous updates as new data emerges. While this study focuses on rainstorm-related BHR in Hong Kong, the ontology structure we propose can be expanded to encompass other disaster contexts, including earthquakes, heatwaves, and pandemics, offering broader applicability in resilience research.

5.3.3. BHR enhancement

Compared with previous experience in disaster reduction and resilience enhancement, the BHR knowledge graph under rainstorms can provide more comprehensive and structured knowledge to guide the efficient improvement of BHR in the future. The benefits are reflected as follows:

- (1) Supporting disaster response and information acquisition. To effectively observe and enhance BHR under disasters, a large amount of data needs to be tracked, such as disaster data, affected building data, impacted occupant data, and emergency resource data. Manually tracking and sharing this data with all stakeholders is challenging. Section 4 of this paper presents a structured knowledge graph of Hong Kong's BHR under rainstorm interference, showing that "debris flow" and "floods" are the most frequent secondary disasters. Therefore, emergency response against these disasters should be prioritized in BHR strategies for rainstorm interference.
- (2) Improving inter-institutional collaboration efficiency. Disaster reduction and BHR enhancement are complex tasks involving multiple stakeholders. The exploration of Hong Kong's BHR knowledge graph subgraph under rainstorm interference in section 5.2 reveals that current disaster response measures of various institutions in Hong Kong have overlapping issues. Closer

cooperative relationships between government departments and other institutions can be established to avoid redundancy in emergency measures and resource waste. Long-term collaborative frameworks will enhance the implementation efficiency of disruption response strategies. The BHR knowledge graph provides clear guidance for stakeholders to better prepare for disasters, thereby enhancing BHR and minimizing disaster losses.

- (3) Optimizing decision-making and strategy formulation. Constructing a BHR ontology and knowledge graph systemizes and structures knowledge. Governments and disaster emergency departments can actively build dynamically updated knowledge graphs for specific disasters, storing and updating data in a structured manner to facilitate efficient query and application in disaster reduction. This enables relevant governance agencies to quickly access historical disaster data, real-time weather conditions, and effectiveness evaluations of emergency response measures.

5.3.4. Limitations and future research directions

This study also presents some limitations. Regarding videos in multi-modal data, only the subtitles' information was utilized. With advancements in video-to-text generative models, further valuable knowledge could be extracted from videos. The LLM-empowered ontology construction method highly relies on tokenization NLP models and domain-specific proprietary noun databases. Future research on using LLMs for domain-specific ontology construction should strictly screen appropriate tokenization models and construct noun databases validated by experts to ensure the accuracy and consistency of generated content. In terms of the BHR knowledge, although this study designed stringent prompts based on ICL and incorporated FCM, the knowledge extraction output still contains a few extraneous elements. Future studies could continue to refine the precision and

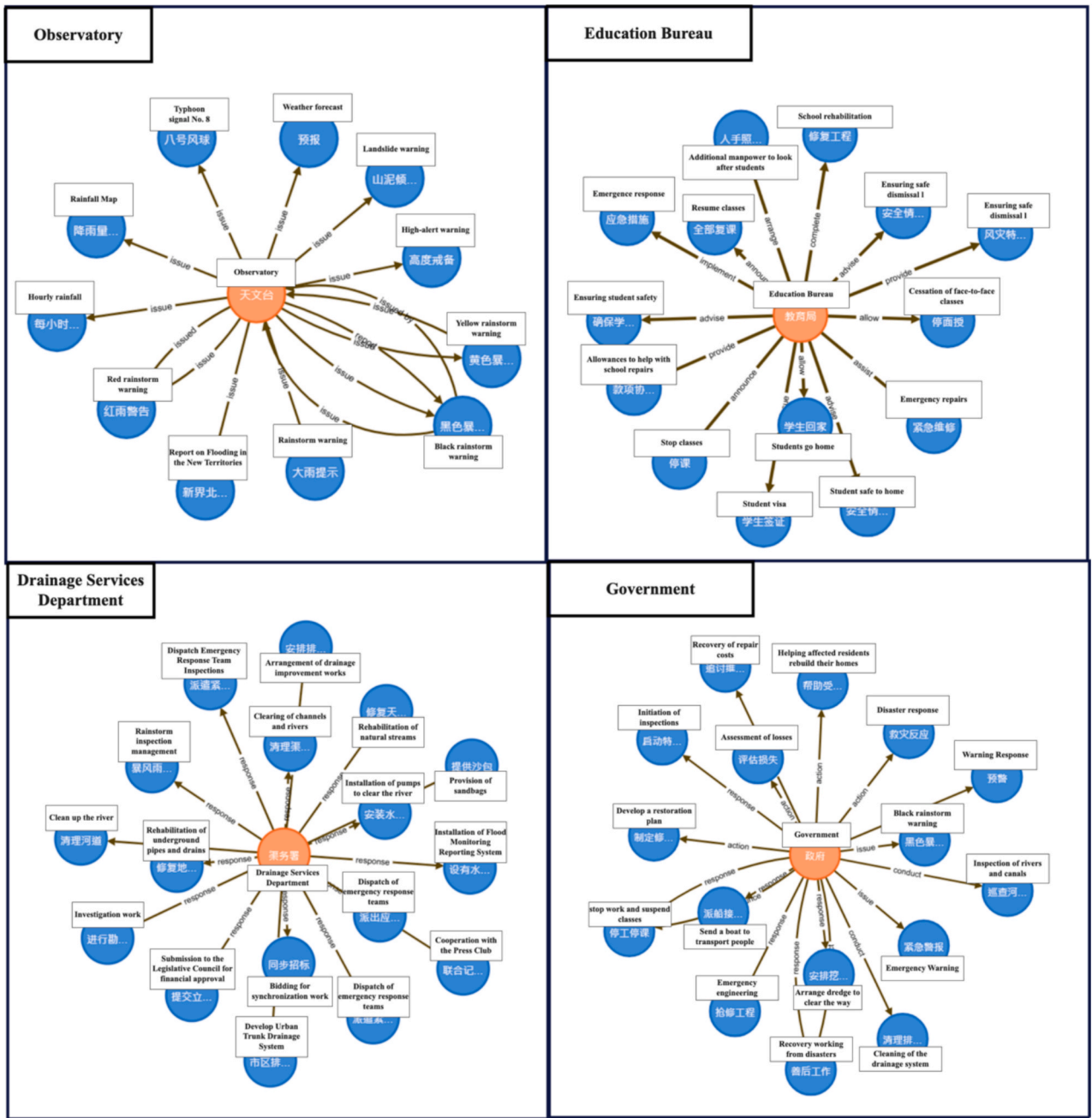


Fig. 14. Subgraphs of organizations and actions in BHR knowledge graph.

accuracy of structured knowledge extraction from the training process of LLMs. In the framework of automated knowledge graph construction empowered LLMs, different LLMs' performance in processes such as ontology construction and entity recognition should be compared, which will make this framework more universal.

What's more, it is possible to delve deeper into automating the knowledge graph construction process with LLMs, such as by training local LLMs and integrating them with other generative AI tools to create a dedicated AI agent for knowledge graph automation. This could lead to the capacity to construct knowledge graphs of all types automatically. For the research on the BHR, more diversified data can also be incorporated, such as Resource Description Framework (RDF) data, Web Ontology Language (OWL) data, and so on. In this way, diversified urban linked data can be obtained. As a result, the knowledge graph will have richer data. With the empowerment of the multi-step workflow of the LLMs, it is possible to screen and calculate among complex data.

Consequently, more issues regarding the health resilience of buildings can be solved, and this can assist in formulating governance strategies for the health resilience of buildings.

6. Conclusion

The study proposes a method utilizing LLMs to empower the automated construction of BHR knowledge graph based on multi-modal data. The structured knowledge in knowledge graph can provide insight into exploring the ontology, influencing factors, relationships, and resilience enhancement strategies of BHR. To enhance the richness of the knowledge graph, this method extends the traditional textual data to multi-modal data, including text, images, and videos, for automated knowledge graph construction. LLMs are employed to deal with multi-modal data and empower the automation construction process at five levels: ontology design, multi-modal data processing, triple extraction,

FCM for triples, and validation of triple extraction. A BHR knowledge graph under Hong Kong rainstorms is constructed as a case study, and the performance of this BHR knowledge graph is also validated by NLP metrics and LLMs-Eval.

This study contributes to developing a novel approach to constructing comprehensive BHR knowledge graphs automatically based on multi-modal data, and also reveals the structured knowledge about BHR under rainstorms or other disasters. The BHR knowledge graph can help fully understand the entire BHR response process and also support the decision-making in guiding the BHR enhancement strategies. In addition, this novel approach not only fits BHR knowledge graph but also can be applied to construct knowledge graph for other new concepts from other research fields.

CRediT authorship contribution statement

Tianlong Shan: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Fan Zhang:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Albert P.C. Chan:** Writing – review & editing, Supervision. **Shiyao Zhu:** Writing – review & editing. **Kaijian Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study is funded by the Start-up Fund for RAPs under the Strategic Hiring Scheme of the Hong Kong Polytechnic University (P0046621) and Research Incentive Scheme of the Hong Kong Polytechnic University (P0048066). Thanks to all researchers and experts who have made great efforts in the study's progress.

Data availability

Data will be made available on request.

References

- [1] F. Zhang, Z. Xu, Y. Yang, M. Qi, H. Zhang, Virtual reality-based evaluation of indoor earthquake safety actions for occupants, *Adv. Eng. Inf.* 49 (2021) 101351, <https://doi.org/10.1016/j.aei.2021.101351>.
- [2] M. Awada, B. Becerik-Gerber, S. Hoque, Z. O'Neill, G. Pedrielli, J. Wen, T. Wu, Ten questions concerning occupant health in buildings during normal operations and extreme events including the COVID-19 pandemic, *Build. Environ.* 188 (2021) 107480, <https://doi.org/10.1016/j.buildenv.2020.107480>.
- [3] F. Zhang, A.P.C. Chan, L. Chen, D. Li, P. Cui, Exploring and measuring the health resilience of urban buildings against the pandemic : a case study of Hong Kong public housing during COVID-19, (2024). DOI: 10.1016/j.jidrr.2024.104343.
- [4] R. Huang, G. Liu, K. Li, Z. Liu, X. Fu, J. Wen, Evolution of residents' cooperative behavior in neighborhood renewal: an agent-based computational approach, *Comput. Environ. Urban Syst.* 105 (2023) 102022, <https://doi.org/10.1016/j.compenvurbsys.2023.102022>.
- [5] T. Shan, F. Zhang, A.P.C. Chan, S. Zhu, K. Li, L. Chen, Y. Wu, Exploring influencing factors of health resilience for urban buildings by integrated CHATGPT-empowered BERTopic model: a case study of Hong Kong, *Environ. Impact Assess. Rev.* 112 (2025) 107852, <https://doi.org/10.1016/j.eiar.2025.107852>.
- [6] M. Zhao, H. Wang, J. Guo, D. Liu, C. Xie, Q. Liu, Z. Cheng, Construction of an Industrial Knowledge Graph for Unstructured Chinese Text Learning, *Appl. Sci.* 9 (2019) 2720, <https://doi.org/10.3390/app9132720>.
- [7] Y. Feng, L. Zhou, Y. Zheng, R. He, C. Ma, Y. Li, Knowledge Graph-based Thought: a knowledge graph enhanced LLMs framework for pan-cancer question answering, (2024) 2024.04.17.589873. DOI: 10.1101/2024.04.17.589873.
- [8] S. Choi, S. Yoon, GPT-based data-driven urban building energy modeling (GPT-UBEM): Concept, methodology, and case studies, *Energy. Buildings* 325 (2024) 115042, <https://doi.org/10.1016/j.enbuild.2024.115042>.
- [9] J. Feng, Y. Du, T. Liu, S. Guo, Y. Lin, Y. Li, CityGPT: Empowering Urban Spatial Cognition of Large Language Models (2024), <https://doi.org/10.48550/arXiv.2406.13948>.
- [10] Y. Sun, X. Ma, Y. Yan, M.-O. Pun, B. Huang, A Novel Automated Urban Building Analysis Framework based on GPT and SAM, in: *IGARSS 2024–2024 IEEE International Geoscience and Remote Sensing Symposium*, 2024, pp. 4666–4669, <https://doi.org/10.1109/IGARSS53475.2024.10640519>.
- [11] J. Song, S. Yoon, Ontology-assisted GPT-based building performance simulation and assessment: implementation of multizone airflow simulation, *Energy. Buildings* 325 (2024) 114983, <https://doi.org/10.1016/j.enbuild.2024.114983>.
- [12] X. Chen, S. Jia, Y. Xiang, A review: Knowledge reasoning over knowledge graph, *Expert Syst. Appl.* 141 (2020) 112948, <https://doi.org/10.1016/j.eswa.2019.112948>.
- [13] L.-I. Wu, Y. Su, G. Li, Zero-Shot Construction of Chinese Medical Knowledge Graph with GPT-3.5-turbo and GPT-4, *ACM Trans. Manage. Inf. Syst.* (2024), <https://doi.org/10.1145/3657305>.
- [14] K. Arrow, B. Bolin, R. Costanza, P. Dasgupta, C. Folke, C.S. Holling, B.-O. Jansson, S. Levin, K.-G. Mäler, C. Perrings, D. Pimentel, Economic growth, carrying capacity, and the environment, *Science* 268 (1995) 520–521, <https://doi.org/10.1126/science.268.5210.520>.
- [15] C.S. Holling, Resilience and stability of ecological systems, *Annu. Rev. Ecol. Syst.* 4 (1973) 1–23, <https://doi.org/10.1146/annurev.es.04.110173.000245>.
- [16] C. Folke, S. Carpenter, B. Walker, M. Scheffer, J. Rockström, Resilience thinking: integrating resilience, adaptability and transformability, *Ecol. Soc.* 15 (2010), <https://doi.org/10.5751/ES-03610-150420>.
- [17] G. Huang, D. Li, X. Zhu, J. Zhu, Influencing factors and their influencing mechanisms on urban resilience in China, *Sustain. Cities Soc.* 74 (2021) 103210, <https://doi.org/10.1016/j.scs.2021.103210>.
- [18] M. Sajjad, J.C.L. Chan, S.S. Chopra, Rethinking disaster resilience in high-density cities: towards an urban resilience knowledge system, *Sustain. Cities Soc.* 69 (2021) 102850, <https://doi.org/10.1016/j.scs.2021.102850>.
- [19] C. Zhai, Y. Zhao, W. Wen, H. Qin, L. Xie, A novel urban seismic resilience assessment method considering the weighting of post-earthquake loss and recovery time, *Int. J. Disaster Risk Reduct.* 84 (2023) 103453, <https://doi.org/10.1016/j.ijdrr.2022.103453>.
- [20] M.Z. Serdar, M. Koç, S.G. Al-Ghamdi, Urban Transportation Networks Resilience: Indicators, Disturbances, and Assessment Methods, *Sustain. Cities Soc.* 76 (2022) 103452, <https://doi.org/10.1016/j.scs.2021.103452>.
- [21] B. Mayer, A Review of the Literature on Community Resilience and disaster Recovery, *Curr. Environ. Health Rpt* 6 (2019) 167–173, <https://doi.org/10.1007/s40572-019-00239-3>.
- [22] R. Djalante, R. Shaw, A. DeWit, Building resilience against biological hazards and pandemics: COVID-19 and its implications for the Sendai Framework, *Prog. Disaster Sci.* 6 (2020) 100080, <https://doi.org/10.1016/j.pdisas.2020.100080>.
- [23] K. Himoto, Conceptual framework for quantifying fire resilience – a new perspective on fire safety performance of buildings, *Fire Saf. J.* 120 (2021) 103052, <https://doi.org/10.1016/j.firesaf.2020.103052>.
- [24] C.Y. Siu, W. O'Brien, M. Touchie, M. Armstrong, A. Laouadi, A. Gaur, Z. Jandaghian, I. Macdonald, Evaluating thermal resilience of building designs using building performance simulation – a review of existing practices, *Build. Environ.* 234 (2023) 110124, <https://doi.org/10.1016/j.buildenv.2023.110124>.
- [25] G. Xu, T. Guo, A. Li, H. Zhang, K. Wang, J. Xu, L. Dang, Seismic resilience enhancement for building structures: a comprehensive review and outlook, *Structures* 59 (2024) 105738, <https://doi.org/10.1016/j.istruc.2023.105738>.
- [26] N.E. Klepeis, W.C. Nelson, W.R. Ott, J.P. Robinson, A.M. Tsang, P. Switzer, J. V. Behar, S.C. Hern, W.H. Engelmann, The National Human Activity Pattern Survey (NHAPS): a resource for assessing exposure to environmental pollutants, *J. Expo. Sci. Environ. Epidemiol.* 11 (2001) 231–252, <https://doi.org/10.1038/sj.jea.7500165>.
- [27] T. Shan, F. Zhang, A.P.C. Chan, S. Zhu, K. Li, Y. Wu, A multi-source exploration of influencing factors of health resilience for urban buildings: Integrated literature-based identification and CHATGPT-empowered BERTopic model, *Sustain. Cities Soc.* (2024).
- [28] M. Uschold, M. Gruninger, Ontologies: principles, methods and applications, *Knowl. Eng. Rev.* 11 (1996) 93–136, <https://doi.org/10.1017/S0269888900007797>.
- [29] H. Rodríguez, S. Climent, P. Vossen, L. Bloksma, W. Peters, A. Alonge, F. Bertagna, A. Roventini, The Top-down strategy for Building EuroWordNet: vocabulary coverage, base concepts and top ontology, *Comput. Hum.* 32 (1998) 117–152, <https://doi.org/10.1023/A:1001169525131>.
- [30] X. Zheng, B. Wang, Y. Zhao, S. Mao, Y. Tang, A knowledge graph method for hazardous chemical management: Ontology design and entity identification, *Neurocomputing* 430 (2021) 104–111, <https://doi.org/10.1016/j.neucom.2020.10.095>.
- [31] S. Singh, S. Ghosh, J. Jayaram, M.K. Tiwari, Enhancing supply chain resilience using ontology-based decision support system, *Int. J. Comput. Integr. Manuf.* 32 (2019) 642–657, <https://doi.org/10.1080/0951192X.2019.1599443>.
- [32] L. van Wassenae, E. Oosterkamp, M. van Asseldonk, M. Ryan, Food system resilience: ontology development and impossible trinitities, *Agric. Food Secur.* 10 (2021) 38, <https://doi.org/10.1186/s40066-021-00332-7>.
- [33] A. Sheth, S. Padhee, A. Gyrard, Knowledge Graphs and Knowledge networks: the story in brief, *IEEE Internet Comput.* 23 (2019) 67–75, <https://doi.org/10.1109/MIC.2019.2928449>.
- [34] T. Al-Moslimi, M. Gallofré Ocaña, A.L. Opdahl, C. Veres, Named Entity extraction for knowledge graphs: a literature overview, *IEEE Access* 8 (2020) 32862–32881, <https://doi.org/10.1109/ACCESS.2020.2973928>.

- [35] V. Yadav, S. Bethard, A Survey on Recent Advances in Named Entity Recognition from Deep Learning models, in: E.M. Bender, L. Derczynski, P. Isabelle (Eds.), Proceedings of the 27th International Conference on Computational Linguistics, Association for Computational Linguistics, Santa Fe, New Mexico, USA, 2018: pp. 2145–2158. <https://aclanthology.org/C18-1182> (accessed July 11, 2024).
- [36] T. Boros, S.D. Dumitrescu, R. Burtica, NLP-Cube: End-to-End Raw Text Processing With Neural Networks, in: D. Zeman, J. Hajič (Eds.), Proceedings of the CoNLL 2018 Shared Task: Multilingual Parsing from Raw Text to Universal Dependencies, Association for Computational Linguistics, Brussels, Belgium, 2018: pp. 171–179. DOI: 10.18653/v1/K18-2017.
- [37] C. Liu, S. Yang, A text mining-based approach for understanding chinese railway incidents caused by electromagnetic interference, Eng. Appl. Artif. Intel. 117 (2023) 105598, <https://doi.org/10.1016/j.engappai.2022.105598>.
- [38] L. Gan, B. Ye, Z. Huang, Y. Xu, Q. Chen, Y. Shu, Knowledge graph construction based on ship collision accident reports to improve maritime traffic safety, Ocean Coast. Manag. 240 (2023) 106660, <https://doi.org/10.1016/j.ocecoaman.2023.106660>.
- [39] W. Fang, L. Ma, P.E.D. Love, H. Luo, L. Ding, A. Zhou, Knowledge graph for identifying hazards on construction sites: Integrating computer vision with ontology, Autom. Constr. 119 (2020) 103310, <https://doi.org/10.1016/j.autcon.2020.103310>.
- [40] J. Liu, F. Schmid, K. Li, W. Zheng, A knowledge graph-based approach for exploring railway operational accidents, Reliab. Eng. Syst. Saf. 207 (2021) 107352, <https://doi.org/10.1016/j.res.2020.107352>.
- [41] Y. Zhu, A knowledge graph and BiLSTM-CRF-enabled intelligent adaptive learning model and its potential application, Alex. Eng. J. 91 (2024) 305–320, <https://doi.org/10.1016/j.aej.2024.02.011>.
- [42] C. Chang, Y. Tang, Y. Long, K. Hu, Y. Li, J. Li, C.-D. Wang, Multi-Information Preprocessing Event Extraction with BiLSTM-CRF attention for Academic Knowledge Graph Construction, IEEE Trans. Comput. Social Syst. 10 (2023) 2713–2724, <https://doi.org/10.1109/TCSS.2022.3183685>.
- [43] H. Xu, G. Fan, G. Kuang, C. Wang, Exploring the potential of BERT-BiLSTM-CRF and the attention mechanism in building a tourism knowledge graph, Electronics 12 (2023) 1010, <https://doi.org/10.3390/electronics12041010>.
- [44] C. Liu, S. Yang, Using text mining to establish knowledge graph from accident/incident reports in risk assessment, Expert Syst. Appl. 207 (2022) 117991, <https://doi.org/10.1016/j.eswa.2022.117991>.
- [45] Y. Zou, Y. Huang, Y. Wang, F. Zhou, Y. Xia, Z. Shen, The Construction of Urban rainstorm disaster event knowledge graph considering evolutionary processes, Water 16 (2024) 942, <https://doi.org/10.3390/w16070942>.
- [46] C. Jia, Y. Shi, Q. Yang, Y. Zhang, Entity Enhanced BERT Pre-training for Chinese NER, in: B. Webber, T. Cohn, Y. He, Y. Liu (Eds.), Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, Online, 2020: pp. 6384–6396. DOI: 10.18653/v1/2020.emnlp-main.518.
- [47] X. Ren, A. El-Kishky, H. Ji, J. Han, Automatic Entity Recognition and Typing in Massive Text Data, in: Proceedings of the 2016 International Conference on Management of Data, Association for Computing Machinery, New York, NY, USA, 2016: pp. 2235–2239. DOI: 10.1145/2882903.2912567.
- [48] K. Li, T. Shan, H. Wu, Z. Zou, R. Huang, R. Chang, A. Shrestha, Technology opportunity discovery linking artificial intelligence and construction technologies: a graph convolution network-based approach, Eng. Appl. Artif. Intel. 138 (2024) 109401, <https://doi.org/10.1016/j.engappai.2024.109401>.
- [49] F. Liu, J. Jiang, Y. Lu, Z. Huang, J. Jiang, The ethical security of large language models: a systematic review, Front. Eng 12 (2025) 128–140, <https://doi.org/10.1007/s42524-025-4082-6>.
- [50] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, D. Zhou, Chain-of-Thought Prompting Elicits Reasoning in Large Language Models (2023), <https://doi.org/10.48550/arXiv.2201.11903>.
- [51] K.I. Roumeliotis, N.D. Tselikas, ChatGPT and Open-AI Models: a Preliminary Review, Future Internet 15 (2023) 192, <https://doi.org/10.3390/fi15060192>.
- [52] J.F. Li, W. Li, X. Wei, W. Xu, Supply Chain Risk Measurement and Resilience: Insights from Large Language Models, (2025). <https://papers.ssrn.com/abstract=5072995> (accessed April 26, 2025).
- [53] X. Li, L. Ding, Y. Zhao, Navigating Turbulence with Large language models and neural network: a machine learning approach to predicting resilience across global automotive supply, Networks (2024), <https://doi.org/10.2139/ssrn.4991288>.
- [54] B. Kereopa-Yorke, Building resilient SMEs: Harnessing large language models for cyber security in Australia, Journal of AI, Robotics & Workplace Automation 3 (2024) 15–27.
- [55] S. Khandelwal, D. Kumar, in: Computational Fact Validation from Knowledge Graph Using Structured and Unstructured Information, in, Association for Computing Machinery, New York, NY, USA, 2020, pp. 204–208, <https://doi.org/10.1145/3371158.3371187>.
- [56] N. Mayerhofer, Constructing Knowledge Graphs From Unstructured Text Using LLMs, (2024). <https://neo4j.com/blog/developer/construct-knowledge-graphs-unstructured-text/> (accessed April 19, 2025).
- [57] G. Hristova, N. Netov, Media Coverage and Public perception of Distance Learning during the COVID-19 Pandemic: a Topic Modeling Approach based on BERTopic, in, IEEE International Conference on Big Data (big Data) 2022 (2022) 2259–2264, <https://doi.org/10.1109/BigData55660.2022.10020466>.
- [58] Apify Store - 2,000+ web scraping and automation tools · Apify, Apify (n.d.). <https://apify.com/store> (accessed July 11, 2024).
- [59] N.-W. Chi, K.-Y. Lin, S.-H. Hsieh, Using ontology-based text classification to assist Job Hazard Analysis, Adv. Eng. Inf. 28 (2014) 381–394, <https://doi.org/10.1016/j.aei.2014.05.001>.
- [60] M. Grootendorst, BERTopic: Neural topic modeling with a class-based TF-IDF procedure, (2022). <http://arxiv.org/abs/2203.05794> (accessed December 14, 2023).
- [61] T. Goyal, J.J. Li, G. Durrett, News Summarization and Evaluation in the Era of GPT-3, (n.d.).
- [62] M.F. Rahman, W. Liu, S.B. Suhaime, S. Thirumuruganathan, N. Zhang, G. Das, Hdbscan, Density Based Clustering over Location Based Services (2016), <https://doi.org/10.48550/arXiv.1602.03730>.
- [63] OpenAI, (n.d.). <https://openai.com/> (accessed July 11, 2024).
- [64] T. Wu, S. He, J. Liu, S. Sun, K. Liu, Q.-L. Han, Y. Tang, A Brief overview of ChatGPT: the history, Status Quo and Potential Future Development, IEEE/CAA JAS 10 (2023) 1122–1136, <https://doi.org/10.1109/JAS.2023.123618>.
- [65] Y. Zhang, S. Sun, X. Gao, Y. Fang, C. Brockett, M. Galley, J. Gao, B. Dolan, Joint Retrieval and Generation Training for Grounded Text Generation, ArXiv (2021). <https://www.semanticscholar.org/paper/Joint-Retrieval-and-Generation-Training-for-Text-Zhang-Sun/f7b755474da275c021f6950590c89f7461b5c274> (accessed January 23, 2024).
- [66] X. Wei, X. Cui, N. Cheng, X. Wang, X. Zhang, S. Huang, P. Xie, J. Xu, Y. Chen, M. Zhang, Y. Jiang, W. Han, ChatIE: Zero-Shot Information Extraction via Chatting with ChatGPT, (2024). DOI: 10.48550/arXiv.2302.10205.
- [67] M.N. Siddiky, M. Rahman, M.F. Hossen, M. Rahman, M. Jaman, Optimizing AI language models: a study of ChatGPT-4 vs, ChatGPT-4o, Preprints. Org (2025), <https://doi.org/10.20944/preprints202502.0066.v1>.
- [68] Z. Fan, C. Chen, CuPe-KG: Cultural perspective-based knowledge graph construction of tourism resources via pretrained language models, Inf. Process. Manag. 61 (2024) 103646, <https://doi.org/10.1016/j.ipm.2024.103646>.
- [69] O. Rubin, J. Herzig, J. Berant, Learning To Retrieve Prompts for In-Context Learning, in: M. Carpuat, M.-C. de Marneffe, I.V. Meza Ruiz (Eds.), Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Association for Computational Linguistics, Seattle, United States, 2022: pp. 2655–2671. DOI: 10.18653/v1/2022.naacl-main.191.
- [70] Y. Wang, G. Gao, J. Zhai, Q. Liu, L. Song, Evolution characteristics of the rainstorm disaster chains in the Guangdong-Hong Kong-Macao Greater Bay Area, China, Nat Hazards 119 (2023) 2011–2032, <https://doi.org/10.1007/s11069-023-06108-5>.
- [71] X. Wu, Y. Zeng, A. Das, S. Jo, T. Zhang, P. Patel, J. Zhang, S.-J. Gao, D. Pratt, Y.-C. Chiu, Y. Huang, reguloGPT: Harnessing GPT for Knowledge Graph Construction of Molecular Regulatory Pathways, (2024) 2024.01.27.577521. DOI: 10.1101/2024.01.27.577521.
- [72] H. Liu, D. Tam, M. Muqeeth, J. Mohta, T. Huang, M. Bansal, C. Raffel, Few-shot parameter-efficient fine-tuning is better and Cheaper than In-Context, Learning (2022), <https://doi.org/10.48550/arXiv.2205.05638>.
- [73] Y. Liu, Y. Wan, L. He, H. Peng, P.S. Yu, KG-BART: Knowledge Graph-Augmented BART for Generative Commonsense Reasoning, (2021). DOI: 10.48550/arXiv.2009.12677.
- [74] K. Papineni, S. Roukos, T. Ward, W.-J. Zhu, in: Bleu: a Method for Automatic Evaluation of Machine Translation, Association for Computational Linguistics, Philadelphia, Pennsylvania, USA, 2002, pp. 311–318, <https://doi.org/10.3115/1073083.1073135>.
- [75] C.-Y. Lin, ROUGE: a Package for Automatic Evaluation of Summaries, in: T.S.B. Out (Ed.), Association for Computational Linguistics, Spain, Barcelona, 2004, pp. 74–81, <https://aclanthology.org/W04-1013> (accessed July 11, 2024).
- [76] S. Banerjee, A. Lavie, METEOR: An Automatic Metric for MT Evaluation with Improved Correlation with Human Judgments, in: J. Goldstein, A. Lavie, C.-Y. Lin, C. Voss (Eds.), Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization, Association for Computational Linguistics, Ann Arbor, Michigan, 2005: pp. 65–72. <https://aclanthology.org/W05-0909> (accessed July 11, 2024).
- [77] T. Zhang, V. Kishore, F. Wu, K.Q. Weinberger, Y. Artzi, Bertscore: Evaluating Text Generation with BERT (2020), <https://doi.org/10.48550/arXiv.1904.09675>.
- [78] J. Fu, S.-K. Ng, Z. Jiang, P. Liu, Gptscore: Evaluate as You Desire (2023), <https://doi.org/10.48550/arXiv.2302.04166>.
- [79] Ö. Bodin, Collaborative environmental governance: achieving collective action in social-ecological systems, Science 357 (2017) eaan1114, <https://doi.org/10.1126/science.aan1114>.
- [80] R.P. Bixler, K. Lieberknecht, S. Atshan, C.P. Zutz, S.M. Richter, J.A. Belaire, Reframing urban governance for resilience implementation: the role of network closure and other insights from a network approach, Cities 103 (2020) 102726, <https://doi.org/10.1016/j.cities.2020.102726>.
- [81] K. Wulff, D. Donato, N. Lurie, What is health resilience and how can we build it? Annu. Rev. Public Health 36 (2015) 361–374, <https://doi.org/10.1146/annurev-publichealth-031914-122829>.
- [82] E. Trist, Referent organizations and the development of inter-organizational domains, Hum. Relat. 36 (1983) 269–284, <https://doi.org/10.1177/001872678303600304>.
- [83] S. Nop, A. Thornton, P. Tranter, Towards effective stakeholder collaboration in building urban resilience in Phnom Penh: opportunities and obstacles, Environ. Dev. Sustain. 25 (2023) 297–320, <https://doi.org/10.1007/s10668-021-02055-y>.

- [84] J. Zhou, J. Li, G. Zheng, Y. Zhu, X. Wang, C. Zhou, GeoViz: A Multi-View Visualization Platform for Spatio-temporal Knowledge Graph, (2024). DOI: 10.48550/arXiv.2405.03697.
- [85] Z. Pan, C. Su, Y. Deng, J. Cheng, Image2Triplets: a computer vision-based explicit relationship extraction framework for updating construction activity knowledge graphs, *Comput. Ind.* 137 (2022) 103610, <https://doi.org/10.1016/j.compind.2022.103610>.
- [86] Y. Suo, F. Ma, L. Zhu, Y. Yang, Knowledge-Enhanced Dual-stream Zero-shot Composed Image Retrieval, in: 2024: pp. 26951–26962. https://openaccess.thecvf.com/content/CVPR2024/html/Suo_Knowledge-Enhanced_Dual-stream_Zero-shot_Composed_Image_Retrieval_CVPR_2024_paper.html (accessed April 19, 2025).